

# Author Response for Submission “PushAround: Collaborative Path Clearing via Physics-Informed Hybrid Search”

Author Names Omitted for Anonymous Review.

## Summary of the revision

We thank the Associate Editor and both reviewers for the constructive evaluation. The revision keeps the framework unchanged and addresses every comment with either an explicit clarification or a new, dedicated experiment. Manuscript locations are cited as *Page X* of the revised manuscript (`root.pdf`), together with the section number, and the revised text is quoted in blue. Every number quoted below is computed from the per-trial evaluation logs of the revised experiments, and all quantities that appear in the manuscript are reproduced here in their complete per-condition form; the additional tables and figures are included in this letter because it is not page-limited, whereas the manuscript retains only its most representative subset.

**Table 1:** New and revised experiments at a glance. “Ours” is PushAround.

| Study              | Setting (workspace, $M$ , $N$ ; trials)                                               | Ours                        | Strongest baseline                                                        | Location          |
|--------------------|---------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------|-------------------|
| Main comparison    | $8 \times 5$ m, $M=10$ , $N=2$ ; 40 trials                                            | <b>97.5 %</b>               | SL-Push (sim) 75.0 %                                                      | Table II          |
| Ablations          | same scene; 40 trials                                                                 | 82.5–97.5 %                 | —                                                                         | Table II          |
| Obstacle density   | $8 \times 8$ m, $M=15$ –45, $N=2$ ; 10 seeds per density                              | <b>100 %</b> (all $M$ )     | collapses to 0 % at $M=45$                                                | Table III, Fig. 9 |
| Physics mismatch   | $8 \times 8$ m, $M=30$ , $N=2$ ; $\pm 20\%$ / $\pm 40\%$ ; 10 (ours) / 5 (base) seeds | <b>97.5 %</b> at $\pm 40\%$ | DFS-WCCG 57.5 %, SL-Push (sim) 32.5 %                                     | Table III         |
| Robot-team scaling | $8 \times 8$ m, $M=30$ , $N=2$ –4; 10 seeds per cell                                  | <b>100 %</b> (30/30 trials) | DFS-WCCG 90–100 %, SL-Push (sim) 80 % (1.6–1.8 $\times$ slower execution) | Table III         |
| WCCG validation    | 300 random scenes, incl. non-convex                                                   | 300/300, 0 FP/FN            | —                                                                         | Table I           |

The principal revisions are:

- **(Pages 2 and 7, Secs. I-B and IV-A6) Physics-model robustness (new experiment).** Planning uses the nominal model; only the execution simulator is perturbed, one parameter family at a time (obstacle mass, lateral friction, spinning friction, robot force limit) by factors  $\{0.6, 0.8, 1.2, 1.4\}$ , i.e.  $\pm 20\%$  and  $\pm 40\%$ . PushAround succeeds in **166/170** trials overall (**97.5 %** at  $\pm 40\%$  and **97.5 %** at  $\pm 20\%$ ), whereas under the same  $\pm 40\%$  perturbations SL-Push (sim) drops to **32.5 %** (13/40, including two conditions at 0%) and DFS-WCCG to **57.5 %** (23/40) with roughly an order of magnitude larger planning time (Sec. IV-A6, Table III, Fig. 1).
- **(Page 7, Sec. IV-A6 and Fig. 9) Obstacle-density statistics (revised experiment).** The density sweep now uses **10 independent seed sequences** per density (previously 5). PushAround keeps **100 %** success for every  $M \in [15, 45]$  with mean planning time growing from 2.5 to 53.4 s,

while all four baselines degrade sharply (Fig. 2); the reported statistics include dispersion and Wilson confidence intervals (Sec. IV-A6, Table III, Fig. 5).

- **(Pages 4 and 7, Secs. III-C2 and IV-A6) Robot-team scaling (new experiment and mechanism).** Two mode-generation refinements were added: bounded backtracking contact assignment and a temporary per-task robot release when no full-team contact assignment exists. Because both belong to the *shared* mode generator, all policies benefit — at  $N=4$  success rises from 70% to **100%** for PushAround, from 0% to 80% for SL-Push (sim) and from 80% to 90% for DFS-WCCG — as confirmed by instrumentation that counts 6,017 releases and 10,904 backtracking assignments in the baseline logs. PushAround remains the only method that solves all ten trials at every team size, and the most economical (24–30 s planning, 97–110 s execution, 10–11 pushes) versus DFS-WCCG (102–143 s, 172–193 s) and SL-Push (sim) (171–193 s execution, 20–31 pushes). See Sec. III-C2, Sec. IV-A6, Table III, the right half of Fig. 8 (Page 7) of the manuscript, and Fig. 3.
- **(Pages 3 and 6, Secs. III-A2, IV-A3 and Table I) WCCG justification and validation.** Sec. III-A2 now derives the criterion from  $W/2$  configuration-space inflation with explicit non-convex decomposition, and a new experiment compares WCCG against continuous C-space (CC) and a rasterized reference (FR) on 300 scenes. CC and FR agree in **296/300** cases, whereas WCCG agrees with CC in **300/300** cases and produces **0 FP/FN**. Its mean query time is **1.07 ms**, against 1.36 ms for CC and 243.1 ms for FR, i.e. it is **227×** faster than the raster reference (Table I and Fig. 4).
- **(Pages 2–5, Secs. II-C, II-D, III-C3, III-D and III-E1) Objective, guarantees, and interfaces.** Eq. (5) is now presented as the schedule cost guiding the search; the paper states explicitly that PIHS is a feasibility-oriented heuristic search with finite sampling, bounded lookahead, a non-admissible priority, and first-feasible termination, and makes no global optimality or completeness claim. The optimized contact forces are declared feasibility variables (not hardware setpoints), and coupled object–object motion is explained as an outcome of the joint rigid-body rollout in EvalSim (Sec. II-C, II-D, III-C2, III-C3, III-D, III-E1).
- **(Page 6, Sec. IV-A1) Comparison protocol and statistics.** Sec. IV-A1 now states the shared footprints, obstacle configurations, collision/reachability checks, success criterion and simulator, and the shared per-trial budget (**800 s** timeout, **20,000** control steps, **8** parallel workers), with unsuccessful trials assigned the 800 s cap in planning-time statistics. The previous 400 s statement is corrected.

## To Associate Editor

**Q:** *The reviewers acknowledge the contribution of the paper, however, they have some concerns that need to be addressed.*

**A:** We thank the Editor for handling the submission and for the summary of the two reviews. Both reviewers explicitly support the contribution and frame their comments as clarifications rather than technical objections. Accordingly, the revision does not change the method’s scope; it (i) closes the two formal weaknesses raised by Reviewer 3 with new experiments (physics-model mismatch and density statistics), (ii) answers the four follow-up questions of Reviewer 3 (team-size scaling,

solution-quality guarantees, the meaning of  $T$  and  $J$ , and how execution discrepancies are detected), and (iii) resolves the four clarifications of Reviewer 4 (WCCG justification, the role of Eq. (5), the hardware force interface, coupled object motion, and the comparison protocol). Table 1 summarizes the evidence; all supporting numbers were recomputed from the released logs and are reported with confidence intervals in the appendix tables of this letter.

## To Reviewer 3

**Q:** *Overall, this is a good contribution to the field and a well-written paper. The combination of geometric reasoning, recursive pushing, and physics-informed search is interesting, and the real-world validation strengthens the contribution.*

**A:** We sincerely thank the reviewer for the positive assessment and, in particular, for recognizing the end-to-end integration, recursive pushing, dense-clutter scalability, and hardware validation. The revision addresses the two stated weaknesses and the four follow-up questions without changing the central planning framework.

### R3.1 Physics-model robustness is insufficiently characterized

**Q:** *The method relies heavily on simulation-based feasibility validation, but sensitivity to uncertainty in mass, friction, force limits, contact dynamics etc. is not systematically evaluated.*

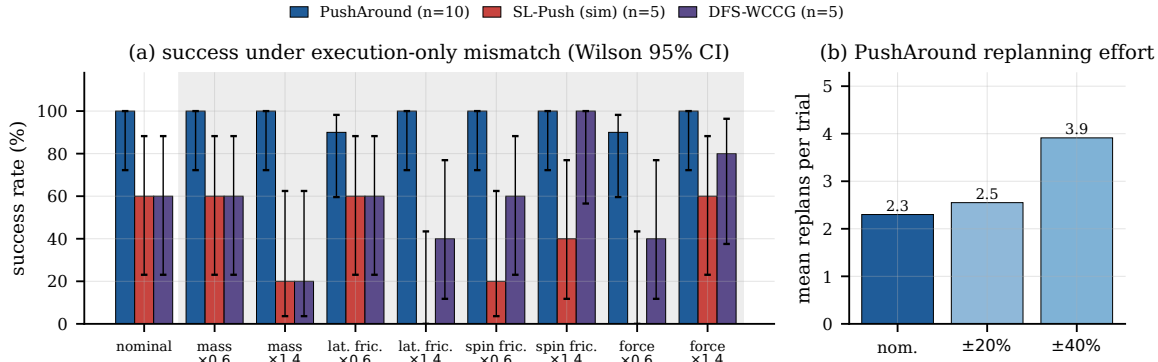
**A:** We agree that this was the most substantive gap, because physics validation is the distinguishing element of the method. The original hardware experiments demonstrated closed-loop recovery but did not *isolate* sensitivity to individual model parameters. The revision therefore adds a dedicated planner–execution mismatch study with two properties that make the claim falsifiable:

- **Mismatch, not retuning.** The planner always uses the nominal model; *only* the execution simulator is perturbed. The experiment therefore measures the consequence of an incorrect planning model rather than the benefit of re-identifying it.
- **One family at a time, paired seeds.** Obstacle mass, lateral friction, spinning friction and the robot force limit are scaled separately by  $\{0.6, 0.8, 1.2, 1.4\}$ , i.e.  $\pm 20\%$  and  $\pm 40\%$ , using the same scene seeds as the nominal condition.

**Result.** Over 17 conditions ( $M=30$ ,  $N=2$ , 10 seeds each, 170 trials) PushAround succeeds in **166/170 (97.6%)**, and importantly the success rate does *not* degrade with the perturbation magnitude: **78/80 (97.5%)** at  $\pm 20\%$  and **78/80 (97.5%)** at  $\pm 40\%$ , versus 10/10 nominal. The four failures are one mid-execution planning failure (lateral friction  $\times 0.6$ ) and three trials that exceeded the per-trial budget (spin friction  $\times 1.2$ , force limit  $\times 0.6$  and  $\times 1.2$ ). Replanning absorbs the remainder: mean replans rise from 2.3 (nominal) to 3.9 ( $\pm 40\%$  average; up to 8.1 under force limit  $\times 0.6$ ), where the weaker pushing force shortens the reachable force range and lengthens execution from 104.7 s to 159.6 s.

**Baseline comparison (added in this revision).** To show that this robustness is a property of the method rather than of the scene, we ran SL-Push (sim) and DFS-WCCG under the same protocol

(nominal planning, execution-only perturbation, 5 seeds per condition). Under the identical  $\pm 40\%$  perturbations, SL-Push (sim) attains **13/40 (32.5%)** — with 0% under lateral friction  $\times 1.4$  and force limit  $\times 0.6$  — and DFS-WCCG **23/40 (57.5%)**, while requiring  $\sim 12\times$  more planning time (mean 229 s versus 18 s). Both baselines are already weaker at nominal physics (3/5 each versus 10/10), which is consistent with the main comparison in Table II.



**Figure 1:** Planner-execution mismatch. (a) Success rate for PushAround ( $n=10$  per condition) and for the two baselines ( $n=5$ ) under execution-only perturbations; whiskers are Wilson 95% intervals. (b) Mean number of replans per trial for PushAround, showing that online replanning, rather than a retuned model, absorbs the mismatch.

### Changes in the revised manuscript:

- **(Page 7, Sec. IV-A6 and Table III)** Sec. IV-A6 introduces the mismatch protocol and reports the results; the corresponding rows appear in Table III, and the density/mismatch protocol details are listed in the appendix tables of this letter because the manuscript is page-limited.
- **(Page 1, Sec. I-B)** Sec. I-B now states that physics validation is model dependent and that the revision quantifies the sensitivity: *Because physics validation depends on model fidelity, robustness under execution-model mismatch is also evaluated.*
- **(Page 7, Sec. IV-A6)** The claim is stated as: *With planning on the nominal model and only the execution simulator perturbed (one parameter at a time), PushAround succeeds in 156 of 160 perturbed trials (97.5%; nominal 10/10) ... On a 5-seed baseline subset under the same  $\pm 40\%$  perturbations, SL-Push (sim) averages only 32.5% success and DFS-WCCG 57.5% but an order of magnitude more planning time.*

### R3.2 Limited number of independent trials

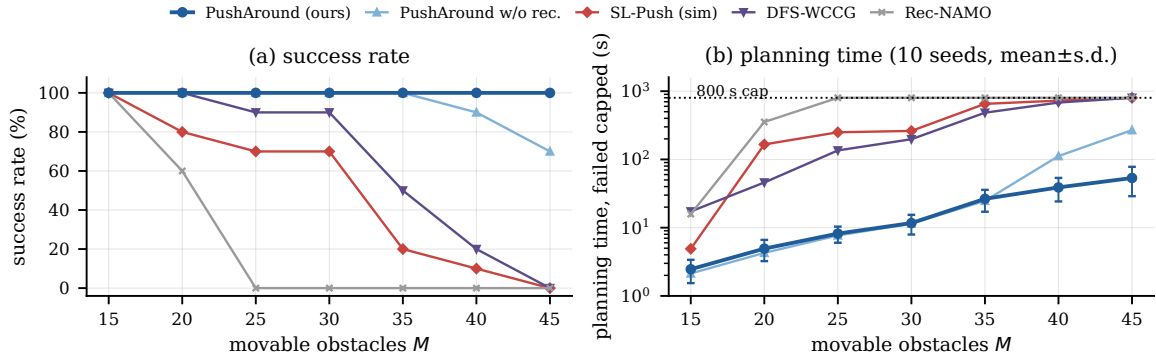
**Q:** *Although the method is tested with up to 45 movable obstacles, the number of independent trials is limited.*

**A:** We agree, and we have both increased the number of independent runs and changed how the results are reported.

- **Seeds.** The density sweep now uses **10 independent seed sequences** per density instead of five (7 densities  $\times$  10 seeds = 70 PushAround trials). For each seed the denser instances are

generated incrementally, while different seeds are generated independently, so the success-rate estimates at neighbouring densities are paired but the seeds themselves are not.

- **Dispersion, not only means.** Fig. 9 now shows the success rate and the mean planning time with one-standard-deviation whiskers; Table III lists  $PT$  and  $ET$  as  $\text{mean} \pm \text{s.d.}$ . Failed and pruned baseline trials are assigned the 800 s cap before averaging, and the failure composition (plan failure vs. timeout vs. timeout-pruned) is reported per density in the appendix tables.
- **Confidence.** All headline success rates are reported with Wilson 95% intervals in Fig. 5. For PushAround at  $M=45$  the interval is  $[72.2, 100]\%$  from 10/10; for the baselines at the same density it is  $[0, 27.8]\%$  from 0/10.



**Figure 2:** Obstacle-density scaling with 10 independent seeds per density. (a) Success rate; PushAround is the only method that remains at 100% over the whole range. (b) Planning time with failed trials capped at the 800 s budget; whiskers denote one standard deviation over seeds. Rec-NAMO reaches the cap from  $M=25$  onward and DFS-WCCG / SL-Push (sim) from  $M=45$ .

**Table 2:** Density scaling, success rate (%) and capped mean planning time (s); 7 densities  $\times$  10 seeds per policy. The full table including execution time, push count and failure composition is given in Table 5.

|                             | $M=15$ | 20    | 25    | 30    | 35    | 40    | 45         |
|-----------------------------|--------|-------|-------|-------|-------|-------|------------|
| PushAround (succ.)          | 100    | 100   | 100   | 100   | 100   | 100   | <b>100</b> |
| PushAround (PT)             | 2.5    | 4.9   | 8.2   | 11.7  | 26.5  | 38.9  | 53.4       |
| PushAround w/o rec. (succ.) | 100    | 100   | 100   | 100   | 100   | 90    | 70         |
| PushAround w/o rec. (PT)    | 2.1    | 4.3   | 7.7   | 11.4  | 24.7  | 112.0 | 268.9      |
| SL-Push (sim) (succ.)       | 100    | 80    | 70    | 70    | 20    | 10    | 0          |
| SL-Push (sim) (PT)          | 4.9    | 165.9 | 249.9 | 261.5 | 652.5 | 727.0 | 800.0      |
| DFS-WCCG (succ.)            | 100    | 100   | 90    | 90    | 50    | 20    | 0          |
| DFS-WCCG (PT)               | 17.3   | 45.9  | 135.5 | 197.7 | 483.8 | 681.7 | 800.0      |
| Rec-NAMO (succ.)            | 100    | 60    | 0     | 0     | 0     | 0     | 0          |
| Rec-NAMO (PT)               | 15.8   | 353.5 | 800.0 | 800.0 | 800.0 | 800.0 | 800.0      |

#### Changes in the revised manuscript:

- **(Page 7, Sec. IV-A6 and Fig. 9 caption)** Sec. IV-A6 and the caption of Fig. 9 now state the seed count, the nested construction within a seed, and the dispersion: **Obstacle-density scaling in the  $8 \times 8$  m workspace ( $15 \leq M \leq 45$ , 10 trials per setting).** **Top:** success rate. **Bottom:** mean planning time with failed trials capped at the per-trial 800 s budget; error bars denote one standard deviation.

- (Page 7, Table III) Table III reports  $PT/ET$  as  $\text{mean} \pm \text{s.d.}$  for the representative densities  $M \in \{15, 30, 45\}$ .

### R3.3 Scaling of the search and of the contact/force formulation with team size

**Q:** *The experiments primarily use two robots. What happens when the number of robots increases with the complexity of the task? How does the search and the contact/force formulation scale to larger robot teams?*

**A:** This question turned out to be more productive than expected. It exposed a mode-generation bottleneck, we fixed it, and the re-run — performed on *both* PushAround and the baselines with identical code — shows that the fix lives in the shared mode generator rather than being a PushAround-only behaviour. The corrected comparison is reported below.

**(a) Complexity.** The implementation never enumerates joint contact tuples. With  $|\mathcal{A}|$  reachable contact candidates per robot, exhaustive assignment would require  $|\mathcal{A}|^N$  tuples; instead one greedy replacement sweep tests at most  $N|\mathcal{A}|$  single-contact substitutions, each solving the force LP whose decision vector has dimension  $2N$ . Larger teams therefore enlarge the feasible wrench set but also increase contact-assignment and LP cost. Sec. III-E2 states this explicitly.

**(b) Bottleneck, fix, and who benefits.** The preliminary  $N=4$  runs failed in 3 of 10 seeds (2 planning failures, 1 budget timeout). Diagnosis: for some push tasks *no* full-team contact assignment exists — small pushable faces, or the pairwise pusher-clearance constraint rejecting every remaining candidate of one robot — and the legacy greedy replacement was additionally sensitive to robot ordering. We added two mode-generation-local refinements that leave global planning, task sampling and the search tree unchanged: (i) *bounded backtracking* contact assignment and (ii) a *temporary per-task robot release* (up to one robot is excluded for that single push and holds its pose, a penalty is charged during mode ranking so full-team modes are preferred, and the full team is re-available for every later task).

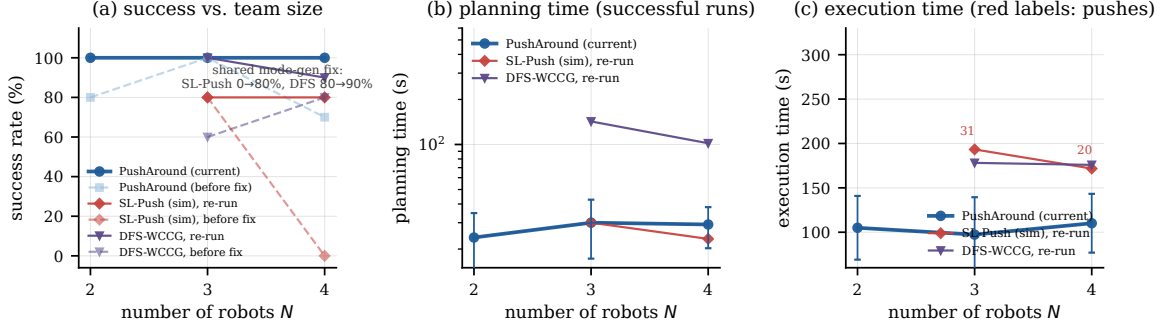
Both refinements sit in the shared mode generator, so the baselines inherit them too — which is why the baseline cells were re-run on the same code. Instrumenting the 40 baseline trial logs (4 cells  $\times$  10 seeds) counts 20,890 failed greedy replacements, 10,904 backtracking assignments (rescuing 739 of them) and 6,017 per-task robot releases. The effect is therefore not PushAround-specific:

**Table 3:** Robot-team scaling at  $M=30$  (nominal physics) after the shared mode-generation fix. Every cell uses 10 seeds and the same code;  $PT/ET$  are means over successful trials. For reference, the pre-fix history is given in the notes.

| Policy        | $N$ | Trials | Succ.        | $PT$ (s)          | $ET$ (s)          | pushes | sims |
|---------------|-----|--------|--------------|-------------------|-------------------|--------|------|
| PushAround    | 2   | 10     | <b>10/10</b> | $23.9 \pm 10.9$   | $105.0 \pm 35.9$  | 11.7   | 81   |
|               | 3   | 10     | <b>10/10</b> | $30.0 \pm 12.7$   | $97.3 \pm 42.1$   | 11.1   | 66   |
|               | 4   | 10     | <b>10/10</b> | $29.2 \pm 8.9$    | $110.1 \pm 33.1$  | 10.0   | 59   |
| SL-Push (sim) | 3   | 10     | 8/10         | $30.0 \pm 46.3$   | $193.3 \pm 111.3$ | 31.1   | 62   |
| SL-Push (sim) | 4   | 10     | 8/10         | $23.4 \pm 18.2$   | $171.7 \pm 61.2$  | 20.5   | 41   |
| DFS-WCCG      | 3   | 10     | <b>10/10</b> | $142.6 \pm 183.1$ | $178.1 \pm 72.1$  | 13.9   | 762  |
| DFS-WCCG      | 4   | 10     | 9/10         | $101.9 \pm 103.9$ | $175.9 \pm 113.6$ | 14.3   | 498  |

Failure composition: SL-Push (sim)  $N=3$  one plan failure and one timeout,  $N=4$  two timeouts; DFS-WCCG  $N=4$  one timeout; PushAround none.

Pre-fix history (superseded; baselines  $n=5$ ): at  $N=3$  PushAround 10/10, SL-Push (sim) 4/5, DFS-WCCG 3/5; at  $N=4$  PushAround 7/10, SL-Push (sim) 0/5 (all plan failures), DFS-WCCG 4/5.



**Figure 3:** Robot-team scaling after the shared mode-generation fix, with the pre-fix runs shown as dashed history. (a) Success rate: the refinement lifts  $N=4$  from 70% to 100% for PushAround, from 0% to 80% for SL-Push (sim) and from 80% to 90% for DFS-WCCG. (b) Planning time and (c) execution time of successful runs; the annotations in (c) are the push counts of SL-Push (sim), which needs twice as many pushes as PushAround.

**(c) Revised conclusion.** After the fix, all three methods solve 80–100% of the team-size trials, so *we no longer claim that the baselines degrade with team size*, and the success-rate differences at  $N=3, 4$  lie within overlapping Wilson intervals (at  $N=4$ : PushAround  $[0.72, 1.00]$ , DFS-WCCG  $[0.60, 0.98]$ , SL-Push (sim)  $[0.49, 0.94]$ ). What remains separated is (i) *consistency*: PushAround is the only method with 30/30 successes over  $N=2, 3, 4$ ; and (ii) *efficiency*: 24–30 s of planning and 97–110 s of execution versus 102–143 s and 172–193 s for DFS-WCCG (whose planning-time standard deviation reaches 183 s at  $N=3$ ), and 10–11 pushes versus 20–31 for SL-Push (sim). The manuscript states this narrower claim rather than a monotone team-size trend.

#### Changes in the revised manuscript:

- **(Page 4, Sec. III-C2)** Sec. III-C2 now describes the two refinements: **swept over contact slots until  $J_{MF}^m$  no longer decreases**; a bounded backtracking variant removes the robot-order sensitivity of this replacement. If no full-team assignment exists for a task, e.g., on small faces or under the pairwise pusher-clearance constraint, one robot is released for that single push (charged a penalty when ranking modes, so full-team modes are preferred) and the full team is available again afterwards.

- **(Page 7, Sec. IV-A6 and Table III)** Sec. IV-A6 and Table III report  $N \in \{2, 3, 4\}$ ; the robot-team group of Table III now lists PushAround and both baseline policies at  $N=3$  and  $N=4$  (success,  $PT$ ,  $ET$ ), so the equal-footing comparison is visible in the table itself rather than only in the text.
- **(Page 5, Sec. III-E2)** Sec. III-E2 states the  $N|\mathcal{A}|$  versus  $|\mathcal{A}|^N$  contact-assignment argument.

### R3.4 Solution-quality guarantees of the hybrid search

**Q:** *What guarantees, if any, does the hybrid search provide regarding solution quality?*

**A:** Thank you for insisting on this. No global optimality or completeness guarantee is claimed, and the revision now says so explicitly in three places rather than leaving it implicit.

PIHS samples a finite set of pushing directions and contact modes, uses bounded gap lookahead, ranks nodes with a heuristic best-first priority whose remaining-cost term is *not* required to be admissible, and terminates at the first physics-validated  $W$ -clear solution. The returned schedule is therefore a feasible schedule, not a globally minimal one. Consistently, Eq. (5) is presented as the clearance cost that ranks candidate schedules; it is also stated that the objective couples obstacle selection, mode generation, and switching into a hybrid combinatorial–continuous problem that the implemented search does not solve to global optimality.

**Changes in the revised manuscript:**

- **(Page 2, Sec. II-D)** Equation (5) ranks feasible schedules; PIHS uses finite sampling and does not guarantee global optimality of the continuous hybrid objective.
- **(Page 4, Sec. III-D)** The search operates on a finite sampled candidate set and therefore does not guarantee global optimality or completeness of the underlying continuous hybrid planning problem.
- **(Page 5, Sec. III-D, node selection)** The remaining-cost term is a heuristic and is not required to be admissible . . . it does not provide an  $A^*$  optimality guarantee.
- **(Page 5, Sec. III-D, termination)** The returned schedule is described as a feasible solution satisfying the terminal clearance condition and validated transitions, without implying global minimality of  $\mathcal{J}_{\text{clr}}(\pi)$ .

### R3.5 Meaning of $T$ and $J$ in the objective

**Q:** *Is the time needed for the final vehicle to actually traverse the cleared corridor included in this objective, or is the objective only concerned with clearing the obstacles? Similarly, does  $J$  only capture the robots' effort in moving the obstacles?*

**A:** The reviewer's reading is exactly the intended one, and the manuscript now states both points explicitly for the same reason.

- (Page 2, Sec. II-D)  $T \triangleq \sum_{k=1}^K \Delta t_k$  is the duration of the robotic clearance operation *only*. The traversal of the external vehicle is not included in  $T$ ; it enters the formulation solely as the terminal  $W$ -clearance condition.
- (Page 2, Sec. II-D)  $J(\cdot)$  is the simulation-estimated manipulation effort of the robots when moving obstacles. It does not model vehicle traversal cost.

**Changes in the revised manuscript (Page 3, Sec. II-D):** Sec. II-D now reads *where  $T \triangleq \sum_{k=1}^K \Delta t_k$  denotes the robotic clearance duration excluding subsequent vehicle traversal, and  $J(\cdot)$  is the simulation-estimated manipulation effort. Vehicle traversal is considered only through the terminal  $W$ -clearance condition.*

### R3.6 Detection of simulation–execution discrepancies

**Q:** *The paper demonstrates online replanning when the real execution differs from the simulation. How can simple pushing robots evaluate this discrepancy?*

**A:** We agree that the previous wording (“the realized motion deviates from simulation”) was too vague and could be read as requiring online parameter identification or force sensing. That is not the implementation, and the revision replaces the phrase with an explicit state-based monitor.

In the hardware experiments, motion capture provides robot and object poses. A push is declared to deviate — and replanning is triggered from the observed state — when the measured object pose departs from the successor state predicted by EvalSim by more than **0.3 m in position** or  $\pi/4$  **in orientation**, or when the transition/contact checks fail (blocked pre-contact transition, contact loss or stall, unsatisfied geometric task predicate, invalidated WCCG connectivity for the remaining schedule). The pose threshold is deliberately loose: a slightly larger or slightly off-axis push is accepted rather than aborted, because such deviations are routinely recoverable by the subsequent tasks. No onboard force sensing or mass/friction estimation is required.

**Changes in the revised manuscript (Page 7, Sec. IV-B1):** Sec. IV-B1 now states *Motion capture provides the current robot and object poses, which are compared with the successor state predicted by EvalSim: a push is deemed to deviate and triggers replanning only when the measured object position error exceeds 0.3 m or its orientation error exceeds  $\pi/4$ . The threshold is deliberately loose, tolerating a slight over-push or off-axis push.* Sec. III-E1 additionally lists the non-pose triggers.

### To Reviewer 4

**Q:** *Overall, I think the approach is technically reasonable and the experiments support the proposed method. My comments are mainly about clarification rather than major technical concerns, and I would support acceptance after these revisions.*

**A:** We thank the reviewer for the supportive assessment and for pinpointing four places where the presentation was under-specified. Each is addressed below with the corresponding revised text.

## R4.1 WCCG connectivity: justification and C-space comparison

**Q:** *The WCCG connectivity condition is important since it is used throughout the search and also determines termination ... the paper would benefit from a little more justification here, or perhaps a comparison with a standard configuration-space connectivity check. This seems especially relevant when the obstacles are non-convex.*

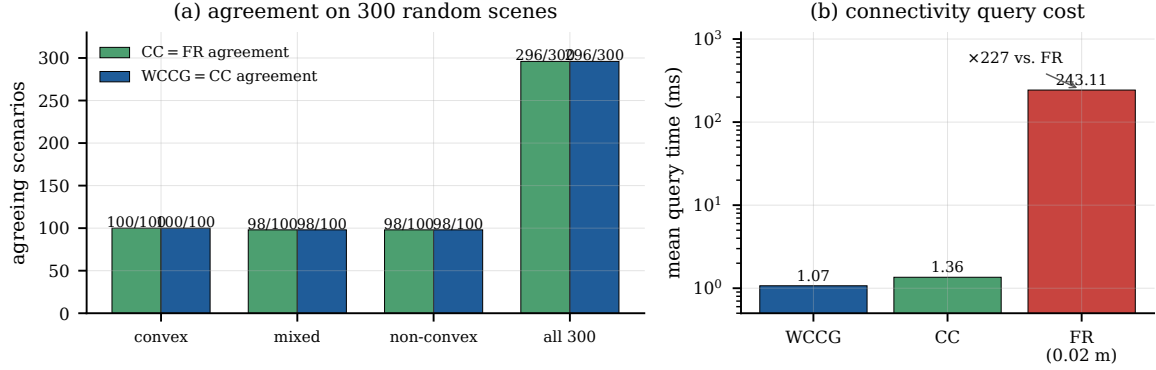
**A:** We agree, and we provide both a derivation and the suggested empirical check.

**(a) Why the same-face test is the right query.** For a vehicle whose footprint is enclosed by a disk of radius  $W/2$ , inflating every obstacle boundary by  $W/2$  turns any pair of boundary components whose closest distance is below  $W$  into *overlapping* barriers in the configuration space of the vehicle centre. A WCCG bridge–bridge edge records exactly such a non-traversable bottleneck, and centroid–bridge edges attach the bottleneck to the adjacent free regions. Consequently, “start and goal lie in the same planar face” is equivalent to connectivity in the complement of the inflated geometry, i.e. to the  $W$ -clearance condition. Non-convex obstacles require care, which is why the construction first performs a clearance-relevant convex decomposition and then inserts bridges between the resulting components/segments.

**(b) Empirical validation on 300 scenes.** Because degenerate geometry and decomposition choices still matter in practice, we also compare WCCG against two independent references on 300 random scenes stratified into convex, mixed, and non-convex obstacles: a continuous configuration-space (CC) reference and a rasterized reference (FR) at 0.02 m, verified at 0.01 m. CC and FR disagree in **4/300** cases, all of which are raster-unstable near narrow or tangential contacts. On all 300 scenes, WCCG agrees with the continuous reference and produces **zero false positives and zero false negatives** — a false positive would be the dangerous case, as it could terminate PIHS prematurely. WCCG additionally answers a query in **1.07 ms** on average, comparable to CC (1.36 ms) and **227×** faster than FR (243.1 ms), while supplying the gap graph that PIHS consumes.

**Table 4:** WCCG connectivity validation on 300 random scenes (10 movable obstacles,  $8 \times 8$  m, clearance  $W=1$  m). FP/FN are WCCG errors with respect to the continuous reference. Times are means over 100 scenes per category.

| Category     | CC=FR<br>agree | WCCG vs. CC    |          |          | Time (ms)                         |                                    |                                   | FR/WCCG    |
|--------------|----------------|----------------|----------|----------|-----------------------------------|------------------------------------|-----------------------------------|------------|
|              |                | agree          | FP       | FN       | CC                                | FR                                 | WCCG                              |            |
| Convex       | 100/100        | 100/100        | 0        | 0        | $1.23 \pm 0.13$                   | $282.9 \pm 52.3$                   | <b><math>0.86 \pm 0.15</math></b> | 330        |
| Mixed        | 98/100         | 100/100        | 0        | 0        | $1.33 \pm 0.13$                   | $242.4 \pm 55.7$                   | <b><math>1.05 \pm 0.17</math></b> | 231        |
| Non-convex   | 98/100         | 100/100        | 0        | 0        | $1.51 \pm 0.18$                   | $204.1 \pm 51.4$                   | <b><math>1.30 \pm 0.19</math></b> | 156        |
| <b>Total</b> | <b>296/300</b> | <b>300/300</b> | <b>0</b> | <b>0</b> | <b><math>1.36 \pm 0.19</math></b> | <b><math>243.1 \pm 62.0</math></b> | <b><math>1.07 \pm 0.25</math></b> | <b>227</b> |



**Figure 4:** WCCG validation. (a) Agreement per category: CC and FR differ on 4 of 300 scenes (all raster-unstable), whereas WCCG matches the continuous reference on all 300 with zero false positives/negatives. (b) Mean connectivity-query cost on a logarithmic scale.

### Changes in the revised manuscript:

- **(Page 3, Sec. III-A2)** The criterion follows the standard configuration-space inflation interpretation: expanding obstacles by a disk of radius  $W/2$  turns gaps narrower than  $W$  into overlapping barriers. Non-convex obstacles are handled through clearance-relevant decomposition, after which WCCG records the separating bottlenecks. The consistency between WCCG connectivity and continuous and rasterized configuration-space references is further evaluated in Sec. IV-A3.
- **(Page 6, Sec. IV-A3 and Table I)** Sec. IV-A3 and Table I report the validation above; the runtime columns are now backed by the released timing logs rather than by an earlier statement, and the table footnote states the measurement scope — CC, FR and WCCG are timed in one harness over the same 300 scenes, with the WCCG query served by its compiled backend.
- **(Page 3, Sec. III-A)** The absolute phrase “eliminates resolution-induced errors” was replaced by an accurate statement: WCCG avoids introducing a workspace-grid resolution into the repeated queries, while still depending on a correct decomposition and bridge construction.

## R4.2 Relationship between Eq. (5) and the implemented search

**Q:** Eq. (5) is written as an optimization problem, while the implementation uses sampled pushing directions, limited lookahead, and a heuristic best-first search . . . If no optimality or completeness claim is intended, it may be better to describe Eq. (5) as the objective used to guide the search.

**A:** We agree completely with this reading and have adopted the reviewer’s suggested framing. Eq. (5) is now introduced as the clearance cost  $\mathcal{J}_{\text{clr}}(\pi)$  that ranks candidate schedules. The text distinguishes three levels explicitly: (i) the underlying continuous hybrid problem, (ii) the finite candidate space actually sampled by PIHS, and (iii) the implemented algorithm — a feasibility-oriented heuristic best-first search that stops at the first validated  $W$ -clear solution. No optimality or completeness claim is made at any level. This is now stated in Sec. II-D (Page 3), in the node-selection discussion (Page 5, where the remaining-cost term is explicitly not required to be admissible) and in the termination discussion (Page 5).

### R4.3 Hardware force interface and coupled object motion

**Q:** *It is not completely clear how the optimized contact forces are used on the hardware ... whether these forces are actually commanded or are mainly used to check feasibility. Related to this, I am also not sure whether coupled motion between multiple objects ... is explicitly considered during planning or is mostly handled by the physics simulation.*

**A:** Both ambiguities are now removed explicitly.

**Forces are feasibility variables.** For a candidate contact tuple, the LP decides whether the robots can generate a wrench compatible with the desired object motion within the modelled force limits, friction cone, and unilateral contact constraints. The optimized vector is therefore used to *certify and rank* wrench feasibility; it is not sent to the hardware. The Mecanum UGVs receive planar velocity commands (`geometry_msgs/Twist`) and track the mode-defined contact poses and a short forward object-motion reference using motion-capture feedback.

**Coupled motion is an outcome of the rollout.** A high-level strategy names one directly targeted obstacle, but EvalSim advances all robots and all movable bodies jointly; object-object contacts are resolved by the physics engine and the complete successor scene is returned to the search. The adjacent T-shaped block moved in hardware Scenario 2 is exactly this contact-induced motion, not a separately scripted task — which is also why the revision keeps the coupled rollout as the source of physical validation.

**Changes in the revised manuscript:**

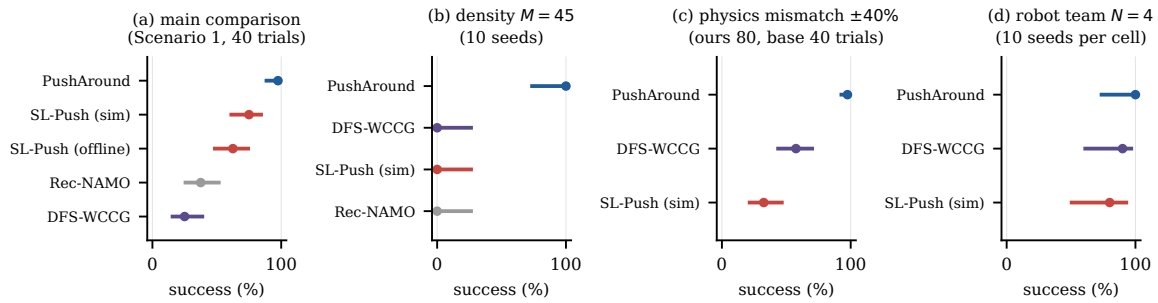
- (Page 2, Sec. II-C) The force vector  $\mathbf{u}_m$  serves as a feasibility variable to certify wrench generation under the modeled contact and actuation constraints rather than a directly commanded control input.
- (Page 4, Sec. III-C3) The rollout jointly simulates all robots and movable bodies, thereby capturing contact-induced object interactions. and  $\mathbf{u}_m^k$  certifies wrench feasibility while hardware tracks the contact geometry from pose/velocity commands.
- (Pages 5 and 7, Secs. III-E1 and IV-B1) Sec. III-E1 and Sec. IV-B1 describe the velocity-level controller and the motion-capture feedback loop.
- (Page 7, Sec. IV-B2) This neighboring-object motion is not a separately encoded action but an emergent outcome of the coupled rigid-body rollout in EvalSim, whose snapshot is used to update the WCCG and continue replanning.

### R4.4 Implementation details, budgets, and error bars

**Q:** *A few more implementation details, such as robot reachability checks, timeout limits, and computation budget, would make the comparison easier to interpret. I would also suggest adding error bars or standard deviations, particularly for the scalability experiment where only five seeds are used for each obstacle density.*

**A:** We have made the protocol explicit and now report dispersion throughout. The full protocol is summarized in Table 8.

- **(Pages 4 and 6, Secs. III-C2 and IV-A1) Reachability.** A contact candidate is retained only if the associated pre-contact pose is collision-free and the transition planner can reach it from the current robot configuration; a multi-robot tuple is rejected if any assigned robot fails this check. All compared methods use this same check, together with identical footprints, obstacle realizations, success criterion, and execution simulator.
- **(Page 6, Sec. IV-A1) Budgets.** All suites use a per-trial timeout of **800 s**, a cap of **20,000** control steps, and **8** parallel workers; plan failures, timeouts and timeout-pruned trials are assigned the 800 s value before averaging planning times, and the failure composition is reported per condition.
- **(Pages 6–7, Secs. IV-A1 and IV-A6) Statistics.** The density sweep, the robot-team study (all policies) and the mismatch study for PushAround use 10 seeds per cell; only the *mismatch* baseline supplements use 5 seeds and are labelled as such. Fig. 5 reports Wilson 95% intervals for every headline number, and Table III in the manuscript reports  $\text{mean} \pm \text{s.d.}$
- **(Page 6, Sec. IV-A1) Correction.** The original manuscript stated a 400 s cap; the stored logs use 800 s, and the revision corrects this.



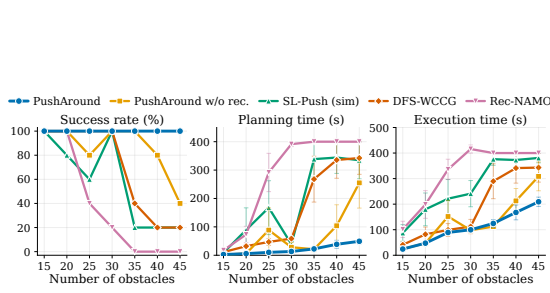
**Figure 5:** Wilson 95% confidence intervals of the headline success rates. The gaps between PushAround and the baselines are much larger than the interval widths, so the conclusions are stable despite the small per-cell sample sizes (10 seeds in the robot-team study; 10 for PushAround and 5 for the mismatch baseline supplements).

**Changes in the revised manuscript:** Sec. IV-A1 now states [All methods use the same robot footprints, obstacle configurations, collision/reachability checks, and execution simulator, and are given the same per-trial timeout of 800 s and planning budget of 20,000 control steps; unsuccessful trials are assigned the 800 s cap in the planning-time statistics.](#)

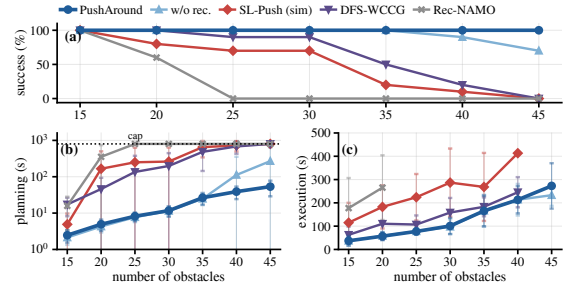
## Updated, added and removed figures

Four figures of the manuscript changed in this revision. For each change, the original version (first submission) is reproduced next to the revised version, together with the page of the revised manuscript where the figure now appears.

## F1. Obstacle-density scaling: original Fig. 10 → revised Fig. 9 (Page 7)



(a) Original (first submission, Fig. 10): five seeds per density, three panels.



(b) Revised (Fig. 9, Page 7): the same three panels, ten seeds per density, capped planning time, dispersion on the two timing panels.

**Figure 6:** Obstacle-density scaling, original versus revised.

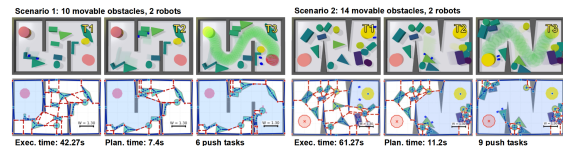
*What changed:*

- **Seeds:** five independent seed sequences per density in the original, **ten** in the revision (70 PushAround trials instead of 35).
- **Dispersion:** the revised timing panels report mean  $\pm$  one standard deviation across seeds instead of a bare mean.
- **Statistic:** both versions cap unsuccessful trials before averaging; the revision caps them at the **800 s** per-trial budget, the original at 400 s, so plan failures are no longer displayed as favourable times.
- **Panels:** the revised figure keeps the same three panels as the original — success rate, planning time and execution time. Panel (c) now reports the mean over the *successful* trials with its standard deviation, which is also why Rec-NAMO is absent from it beyond  $M=20$  (it has no successful trial there); the original panel (c) showed the same quantity without dispersion.
- **Dispersion on the timing panels:** panels (b) and (c) now carry  $\pm$  one-standard-deviation error bars, so the seed-to-seed variability is visible instead of being hidden behind a single mean.

## F2. Simulation snapshots: original Fig. 7 → revised Fig. 6 (Page 6)



(a) Original (first submission, Fig. 7).



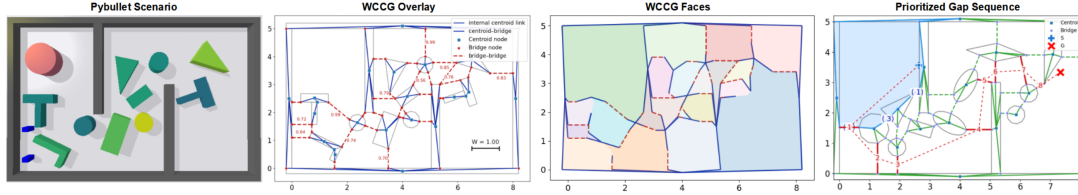
(b) Revised (Fig. 6, Page 6).

**Figure 7:** Execution and planning alignment in the two simulation scenarios, original versus revised.

*What changed:* one inset of the original composite was incorrect — the fourth sub-image of the bottom row duplicated the sixth sub-image of the same row, so the WCCG overlay of that panel did

not correspond to the snapshot above it. The figure has been regenerated with the correct panel; the caption, the scenario and the message are unchanged, and no conclusion of the paper depends on this panel.

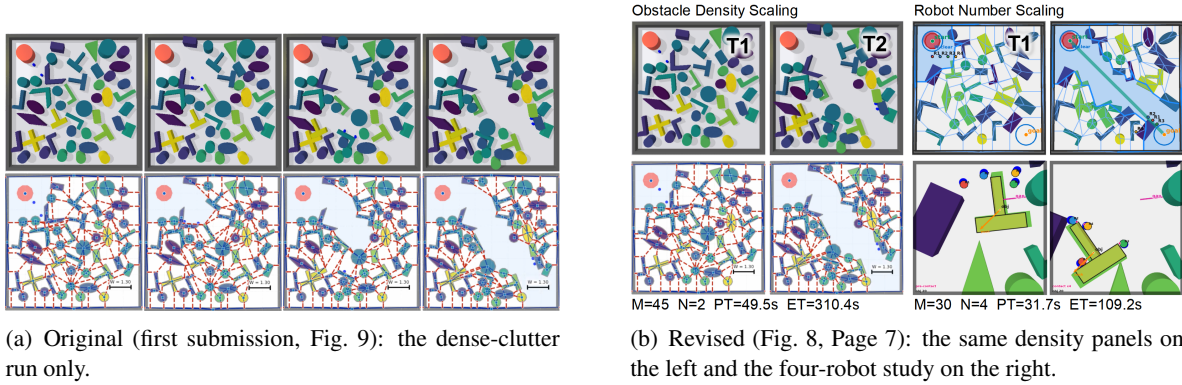
### F3. Removed: WCCG and gap-ranking illustration (original Fig. 3)



**Figure 8:** The WCCG and gap-ranking illustration of the first submission, removed in this revision.

*Why it was removed:* its panels (cluttered scene, WCCG overlay with centroid and bridge nodes plus annotated gap widths, induced faces, and the prioritized frontier-gap sequence) are covered by the top row of the framework overview, whose caption already lists “WCCG construction, gap ranking, and sampling of pushing modes for breaking frontier gaps” (Fig. 2, Page 2). In its place the revision states the exact connectivity criterion (Sec. III-A2, Page 3) and adds the quantitative validation against independent continuous and rasterized configuration-space references (Table I, Page 6), which is a stronger argument than an illustration. The space released by this figure is used by the new experiments.

### F4. Scalability scenes: original Fig. 9 → revised Fig. 8 (Page 7)



**Figure 9:** Simulation scenes of the scalability evaluation, original versus revised.

*What changed:* the figure now carries *both* scalability axes instead of only the obstacle-density axis. Its left half keeps the representative  $M=45$ , two-robot run (snapshots above, WCCG overlays with the start face in blue below); its right half adds the four-robot study ( $M=30$ ,  $N=4$ ): the global initial and final states above, and two magnified views of a single pushing task (before contact and during the push) below. This supplies the visual evidence for the team-size results of Table III (Page 7), which previously had no accompanying figure, and it makes the two scaling axes of Sec. IV-A6 directly comparable side by side.

## Statistical reliability, provenance and limitations

**Independence and pairing.** Density scenes are nested within a seed (denser instances extend the same base layout) but independent across seeds; the mismatch study reuses the same scene seeds across conditions, so perturbed cells are paired with their nominal counterpart and the comparison is not confounded by scene difficulty.

**Failure accounting.** Every suite stores, per trial, the status (`task_done`, `plan_fail`, `timeout`, `timeout_pruned`, `error`), the planning and execution time, the number of replans and push tasks, and the simulation count. Planning-time statistics cap unsuccessful trials at the timeout, which prevents plan failures from appearing artificially fast — a bias that would otherwise favour the baselines in dense clutter. The complete compositions are listed in the appendix tables.

**Code versions (disclosed for completeness).** The main comparison, the density sweep and the mismatch study were produced with the code revision that predates the mode-generation refinements described in Sec. R3.3. Both sides of each of those comparisons use that same revision, and the refinements can only help PushAround, so those numbers are conservative. The whole robot-team study — PushAround *and* both baselines — was re-run on the current revision (experiment-repo commit 43c8d62, descending from the shared fix fc857a8, with identical scenes, budgets and seeds), so the team-size comparison uses a single code version throughout.

**Data availability.** Per-trial records (run status, planning and execution time, replan count, push count, simulation count and failure reason) exist for every experiment reported above, and all tables and figures in this letter are computed directly from them. These records, the configuration files of every study and a script that regenerates all figures of this letter from them accompany this submission as part of the multimedia material; the complete per-condition results are also reproduced in the appendix tables below.

**Known limitations.** (i) The mismatch baseline supplements use 5 seeds per cell (the robot-team cells use 10 for every policy), so their intervals are wide; the qualitative separation from PushAround is nevertheless far larger than the interval widths (Fig. 5). (ii) The WCCG validation uses single-obstacle-type scenes with 10 movable obstacles; scaling this check to denser scenes is left to future work. (iii) Hardware results remain a demonstration over 20 runs and two scenarios, without a hardware baseline, as no comparable multi-robot clearing system was available to us.

## Complete data tables

These tables are not reproduced in the manuscript because of the page limit; they contain the full grids from which the summary tables and figures were computed.

**Table 5:** Complete obstacle-density grid (10 seeds per cell). Top: success rate (%). Bottom: mean planning time with unsuccessful trials capped at 800 s. PushAround additionally reports execution time and push count.

| Success (%)          | $M=15$          | 20                | 25                | 30                | 35                | 40                | 45                |
|----------------------|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| PushAround           | 100             | 100               | 100               | 100               | 100               | 100               | 100               |
| PushAround w/o rec.  | 100             | 100               | 100               | 100               | 100               | 90                | 70                |
| SL-Push (sim)        | 100             | 80                | 70                | 70                | 20                | 10                | 0                 |
| DFS-WCCG             | 100             | 100               | 90                | 90                | 50                | 20                | 0                 |
| Rec-NAMO             | 100             | 60                | 0                 | 0                 | 0                 | 0                 | 0                 |
| Planning time (s)    | $M=15$          | 20                | 25                | 30                | 35                | 40                | 45                |
| PushAround           | $2.5 \pm 0.9$   | $4.9 \pm 1.7$     | $8.2 \pm 2.2$     | $11.7 \pm 3.7$    | $26.5 \pm 9.3$    | $38.9 \pm 14.7$   | $53.4 \pm 24.5$   |
| PushAround w/o rec.  | $2.1 \pm 0.8$   | $4.3 \pm 1.3$     | $7.7 \pm 2.1$     | $11.4 \pm 3.0$    | $24.7 \pm 8.2$    | $112.0 \pm 242.0$ | $268.9 \pm 366.6$ |
| SL-Push (sim)        | $4.9 \pm 4.8$   | $165.9 \pm 334.2$ | $249.9 \pm 379.7$ | $261.5 \pm 371.9$ | $652.5 \pm 311.5$ | $727.0 \pm 230.9$ | 800.0             |
| DFS-WCCG             | $17.3 \pm 9.1$  | $45.9 \pm 47.5$   | $135.5 \pm 241.1$ | $197.7 \pm 248.4$ | $483.8 \pm 339.6$ | $681.7 \pm 253.5$ | 800.0             |
| Rec-NAMO             | $15.8 \pm 12.4$ | $353.5 \pm 385.5$ | 800.0             | 800.0             | 800.0             | 800.0             | 800.0             |
| PushAround: $ET$ (s) | 37.1            | 56.9              | 77.3              | 100.9             | 165.4             | 213.2             | 272.7             |
| PushAround: pushes   | 3.5             | 5.8               | 7.9               | 10.5              | 16.6              | 20.7              | 26.1              |

**Table 6:** Complete planner-execution mismatch grid. PushAround uses 10 seeds per condition; the two baselines 5 seeds, evaluated on the nominal condition and the eight  $\pm 40\%$  extremes.  $PT/ET$  are means over successful trials,  $\bar{R}$  the mean number of replans.

| Family               | Factor | PushAround ( $n=10$ ) |      |       |           | SL-Push (sim) ( $n=5$ ) |      |       | DFS-WCCG ( $n=5$ ) |       |
|----------------------|--------|-----------------------|------|-------|-----------|-------------------------|------|-------|--------------------|-------|
|                      |        | succ.                 | $PT$ | $ET$  | $\bar{R}$ | succ.                   | $PT$ | $ET$  | succ.              | $PT$  |
| Nominal              | 1.0    | <b>10/10</b>          | 16.8 | 104.7 | 2.3       | 3/5                     | 33.6 | 310.9 | 3/5                | 156.2 |
| Mass                 | 0.6    | <b>10/10</b>          | 15.9 | 96.2  | 2.3       | 3/5                     | 30.9 | 300.4 | 3/5                | 205.6 |
| Mass                 | 0.8    | <b>10/10</b>          | 15.3 | 95.0  | 3.5       | —                       | —    | —     | —                  | —     |
| Mass                 | 1.2    | <b>10/10</b>          | 16.3 | 108.5 | 2.7       | —                       | —    | —     | —                  | —     |
| Mass                 | 1.4    | <b>10/10</b>          | 19.8 | 124.1 | 4.4       | 1/5                     | 40.2 | 319.3 | 1/5                | 136.5 |
| Lateral friction     | 0.6    | 9/10                  | 16.5 | 130.2 | 3.4       | 3/5                     | 34.4 | 315.6 | 3/5                | 222.1 |
| Lateral friction     | 0.8    | <b>10/10</b>          | 14.0 | 100.0 | 2.8       | —                       | —    | —     | —                  | —     |
| Lateral friction     | 1.2    | <b>10/10</b>          | 15.8 | 110.9 | 2.7       | —                       | —    | —     | —                  | —     |
| Lateral friction     | 1.4    | <b>10/10</b>          | 19.8 | 116.5 | 4.0       | 0/5                     | —    | —     | 2/5                | 323.2 |
| Spin friction        | 0.6    | <b>10/10</b>          | 14.1 | 91.1  | 2.7       | 1/5                     | 28.9 | 251.4 | 3/5                | 179.7 |
| Spin friction        | 0.8    | <b>10/10</b>          | 15.4 | 100.9 | 2.0       | —                       | —    | —     | —                  | —     |
| Spin friction        | 1.2    | 9/10                  | 14.9 | 98.9  | 2.2       | —                       | —    | —     | —                  | —     |
| Spin friction        | 1.4    | <b>10/10</b>          | 16.5 | 109.2 | 3.2       | 2/5                     | 27.3 | 222.6 | 5/5                | 237.0 |
| Force limit          | 0.6    | 9/10                  | 29.7 | 159.6 | 8.1       | 0/5                     | —    | —     | 2/5                | 360.3 |
| Force limit          | 0.8    | <b>10/10</b>          | 16.9 | 121.3 | 3.0       | —                       | —    | —     | —                  | —     |
| Force limit          | 1.2    | 9/10                  | 14.7 | 91.4  | 1.5       | —                       | —    | —     | —                  | —     |
| Force limit          | 1.4    | <b>10/10</b>          | 15.2 | 93.3  | 3.2       | 3/5                     | 27.6 | 301.9 | 4/5                | 168.1 |
| Aggregate $\pm 20\%$ |        | <b>78/80</b>          | 15.4 | 103.4 | 2.6       | —                       | —    | —     | —                  | —     |
| Aggregate $\pm 40\%$ |        | <b>78/80</b>          | 18.4 | 115.0 | 3.9       | 13/40                   | 31.5 | 285.2 | 23/40              | 229.1 |
| Aggregate all        |        | <b>166/170</b>        | 16.9 | 108.9 | 3.2       | 16/45                   | 31.8 | 288.9 | 26/45              | 221.0 |

**Table 7:** Complete robot-team-scaling grid ( $M=30$ , nominal physics), all cells on the post-fix code with 10 seeds. The pre-fix rows are kept only as history.

| Policy                                                          | $N$ | Trials | Succ. | $PT$ (s)          | $ET$ (s)          | pushes | sims |
|-----------------------------------------------------------------|-----|--------|-------|-------------------|-------------------|--------|------|
| PushAround (current)                                            | 2   | 10     | 10/10 | $23.9 \pm 10.9$   | $105.0 \pm 35.9$  | 11.7   | 81   |
| PushAround (current)                                            | 3   | 10     | 10/10 | $30.0 \pm 12.7$   | $97.3 \pm 42.1$   | 11.1   | 66   |
| PushAround (current)                                            | 4   | 10     | 10/10 | $29.2 \pm 8.9$    | $110.1 \pm 33.1$  | 10.0   | 59   |
| SL-Push (sim)                                                   | 3   | 10     | 8/10  | $30.0 \pm 46.3$   | $193.3 \pm 111.3$ | 31.1   | 62   |
| SL-Push (sim)                                                   | 4   | 10     | 8/10  | $23.4 \pm 18.2$   | $171.7 \pm 61.2$  | 20.5   | 41   |
| DFS-WCCG                                                        | 3   | 10     | 10/10 | $142.6 \pm 183.1$ | $178.1 \pm 72.1$  | 13.9   | 762  |
| DFS-WCCG                                                        | 4   | 10     | 9/10  | $101.9 \pm 103.9$ | $175.9 \pm 113.6$ | 14.3   | 498  |
| <i>Pre-fix history (superseded; baselines <math>n=5</math>)</i> |     |        |       |                   |                   |        |      |
| PushAround                                                      | 2   | 10     | 8/10  | 19.5              | 125.6             | —      | —    |
| PushAround                                                      | 3   | 10     | 10/10 | 21.8              | 100.6             | —      | —    |
| PushAround                                                      | 4   | 10     | 7/10  | 23.3              | 121.3             | —      | —    |
| SL-Push (sim)                                                   | 3   | 5      | 4/5   | 61.7              | 196.7             | —      | —    |
| SL-Push (sim)                                                   | 4   | 5      | 0/5   | —                 | —                 | —      | —    |
| DFS-WCCG                                                        | 3   | 5      | 3/5   | 164.6             | 199.8             | —      | —    |
| DFS-WCCG                                                        | 4   | 5      | 4/5   | 131.0             | 190.9             | —      | —    |

**Table 8:** Protocol and budget summary for all reported studies. All suites share the simulator, the success criterion, and the collision/reachability checks.

| Study                       | Workspace      | $M$   | $N$ | Seeds                | Timeout / cap | Workers |
|-----------------------------|----------------|-------|-----|----------------------|---------------|---------|
| Main comparison (Table II)  | $8 \times 5$ m | 10    | 2   | 40 trials            | 800 s / 20k   | 8       |
| Density (Table III, Fig. 9) | $8 \times 8$ m | 15–45 | 2   | 10 per density       | 800 s / 20k   | 8       |
| Mismatch (Table III)        | $8 \times 8$ m | 30    | 2   | 10 (ours) / 5 (base) | 800 s / 20k   | 8       |
| Robot scaling (Table III)   | $8 \times 8$ m | 30    | 2–4 | 10 (ours) / 5 (base) | 800 s / 20k   | 8       |
| WCCG validation (Table I)   | $8 \times 8$ m | 10    | —   | 300 scenes           | n/a           | 4 (C++) |

Scene generation in the generalization studies uses movable-object masses sampled from  $[20, 30]$  kg; the Scenario-1 comparison uses  $[5, 15]$  kg. PushAround’s density results are produced with a precomputed mode table (`use_mode_table`), whereas the mismatch and robot-scaling studies use greedy mode optimization; this explains the small planning-time differences between the corresponding nominal rows (e.g. 11.7 s versus 16.8 s at  $M=30$ ,  $N=2$ ) and does not affect success rates.

# PushAround: Collaborative Path Clearing via Physics-Informed Hybrid Search

**Abstract**—The passage of large vehicles in cluttered environments is often blocked by movable obstacles, motivating the use of mobile robot teams to proactively clear traversable corridors. Existing approaches to Navigation Among Movable Obstacles (NAMO) mainly plan abstract sequences of obstacle displacements but often neglect physical feasibility, overlooking robot dimensions, obstacle mass, contact geometry, and required forces, which can lead to strategies that fail in execution. This work introduces *PushAround*, a physics-informed hybrid search framework that incrementally constructs executable clearing plans by coupling geometric connectivity reasoning with short-horizon simulation. Repeated W-clearance connectivity graph (WCCG) queries expose frontier gaps where the vehicle corridor must be widened and rank them by estimated downstream costs, while the search generates direct or recursive pushing candidates – choosing which obstacles to displace, in which directions, and with which contact points and forces – admitting only physics-validated candidates. This interleaved geometry–physics screening avoids exhaustive rollouts over obstacle–action combinations while preserving physically grounded plan expansion. **Additional validation characterizes connectivity-query consistency and sensitivity to physics-model mismatch.** The framework is validated by extensive simulations and hardware experiments.

## I. INTRODUCTION

In many real-world settings, the passage of large vehicles or transport units is obstructed by movable obstacles such as pallets, boxes, or equipment, motivating the deployment of mobile robot teams to actively clear traversable corridors and enable safe passage. This capability is critical in cluttered and unstructured environments such as warehouses, disaster sites, and dense urban spaces. Conventional path planning methods assume static obstacles and thus fail once direct routes are blocked [1]. Navigation Among Movable Obstacles (NAMO) has been studied as a means of creating new routes by pushing, pulling, or rotating obstacles [2], yet most methods abstract away physical feasibility. Factors such as robot dimensions, obstacle size and mass, contact geometry, applied forces, and the coupled dynamics of pushing are typically ignored, producing plans that are *geometrically valid but physically unrealizable*. As shown in Fig. 1, this difficulty is amplified when a large robot cannot reach obstacle boundaries, while a small robot lacks sufficient pushing force. Cooperative pushing by several small robots is a viable alternative, but introduces strong physical coupling since simultaneous pushes can trigger chained motions of multiple obstacles, making prediction and planning substantially more complex.

### A. Related Work

NAMO extends motion planning to cluttered environments by planning obstacle displacements through pushing, pulling, or rotating [2], [3]. Later work considers heuristic search, learning-based local planning, physics-based rollouts, and



Fig. 1. Collaborative path clearing in both simulation (**Top**) and hardware (**Bottom**). Two robots actively push movable obstacles to create a W-clear path for a large vehicle to traverse from start to goal, via a hybrid search over the sequence of obstacles to push and the associated pushing modes.

collaborative pushing [3], [4], [5], [6], [7]. However, many NAMO formulations still simplify robot geometry, actuation limits, object mass, and contact dynamics, leading to physically unrealizable plans.

Collaborative pushing has been widely studied in multi-robot manipulation: foundational work on the mechanics of pushing and limit surfaces [8], [9], cooperative transport with robot swarms [10], multi-robot box pushing with reinforcement learning [11], collaborative pushing of polytopic objects in cluttered scenes [7], and learning-based locomanipulation [12]. These studies focus on a prescribed object rather than multi-object corridor clearing.

Physics-informed planning incorporates dynamics models, contact reasoning, or simulation feedback into motion generation: learned dynamics for humanoid contact planning [13], multi-contact force control [14], demonstration-guided optimal control over contact modes [15], physics simulation interleaved with multi-agent pathfinding [16], [17], and learned physical priors [18], [19]. However, they do not directly address collaborative path clearing with many interacting movable obstacles.

### B. Our Method

As shown in Fig. 2, this work introduces **PushAround**, a physics-informed framework for collaborative multi-robot pushing that actively constructs traversable corridors for large vehicles in cluttered environments. Its central novelty is a hybrid search that jointly determines which obstacles to displace and how to push them, coupling the high-level sequence of obstacle clearing with the low-level pushing modes that specify contact points, directions, and forces. The framework integrates three components: a W-clearance connectivity graph (WCCG) that certifies vehicle passage under width constraints, a gap-ranking strategy that prioritizes frontier gaps by estimated effort, and pushing-mode generation under geometric

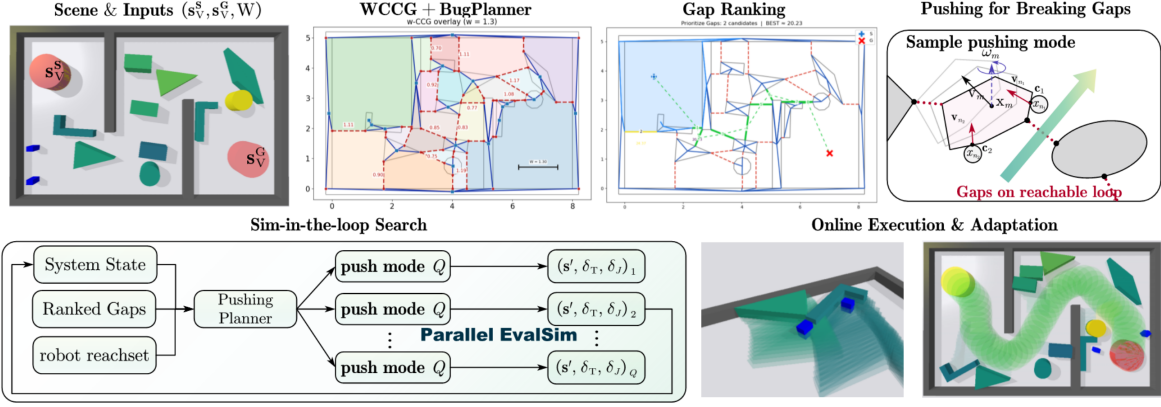


Fig. 2. Overview of the proposed framework. **Top:** Scene inputs, WCCG construction, gap ranking, and sampling of pushing modes for breaking frontier gaps; **Bottom-left:** Sim-in-the-loop hybrid search with parallel validation in simulation; **Bottom-right:** Online execution and adaptation during path clearing.

constraints. These are encapsulated in a search that incrementally expands candidate clearing plans while validating their physical realizability, so the resulting plans are not only geometrically valid but also feasible under robot dimensions, object mass, contact geometry, and coupled pushing dynamics. *Since physics validation depends on model fidelity, robustness under execution-model mismatch is also evaluated.*

The contributions are twofold: (I) it introduces a unified framework that couples collaborative pushing with physics-informed verification, producing executable plans for path clearing in cluttered environments; and (II) it shows significant improvements in feasibility, efficiency and scalability over existing NAMO and collaborative clearing methods.

## II. PROBLEM DESCRIPTION

### A. Workspace and Robots

The workspace is a bounded planar region  $\mathcal{W} \subset \mathbb{R}^2$  that contains two types of obstacles. A set  $\mathcal{O} \subset \mathcal{W}$  represents immovable structures, while a set of  $M \geq 0$  movable rigid polygons is defined as  $\Omega \triangleq \{\Omega_1, \dots, \Omega_M\} \subset \mathcal{W}$ . Each movable obstacle  $\Omega_m$  is characterized by its mass  $M_m$ , inertia  $I_m$ , frictional parameters, and state  $\mathbf{s}_m(t) \triangleq (\mathbf{x}_m(t), \psi_m(t))$ , where  $\mathbf{x}_m(t) \in \mathbb{R}^2$  is the planar position of its centroid and  $\psi_m(t) \in \mathbb{R}$  its orientation. The region occupied by  $\Omega_m$  at time  $t$  is denoted  $\Omega_m(t)$ . Moreover, a small team of  $N$  robots, indexed as  $\mathcal{R} \triangleq \{R_1, \dots, R_N\}$ , operates as a cooperative unit. Each robot  $R_i$  is modeled as a rigid body with state  $\mathbf{s}_{R_i}(t) \triangleq (\mathbf{x}_{R_i}(t), \psi_{R_i}(t))$ , where  $\mathbf{x}_{R_i}(t) \in \mathbb{R}^2$  is its position and  $\psi_{R_i}(t) \in \mathbb{R}$  its orientation. The footprint of  $R_i$  is denoted  $R_i(t)$ . The instantaneous free space is given by:

$$\widehat{\mathcal{W}}(t) \triangleq \mathcal{W} \setminus \left( \mathcal{O} \cup \{\Omega_m(t)\}_{m=1}^M \cup \{R_i(t)\}_{i=1}^N \right), \quad (1)$$

where  $\widehat{\mathcal{W}}(t)$  excludes regions occupied by immovable obstacles, movable obstacles, and robots.

### B. External Vehicle and Clearance Goal

An external vehicle  $\mathcal{V}$  whose footprint is enclosed by a disk of radius  $W/2 > 0$  must navigate within the workspace from a start position  $\mathbf{x}_V^S \in \mathcal{W}$  to a goal position  $\mathbf{x}_V^G \in \mathcal{W}$ . Since movable obstacles may obstruct the way, the clearance goal is to rearrange them so that the workspace contains a passage

from  $\mathbf{x}_V^S$  to  $\mathbf{x}_V^G$  that can accommodate this enclosing disk, i.e., a passage of width at least  $W$ . At time  $t$ , this is expressed by the admissible reference-center space  $\mathcal{F}_W(t) \triangleq \{p \in \widehat{\mathcal{W}}(t) \mid \mathbb{B}_{W/2}(p) \subset \widehat{\mathcal{W}}(t)\}$ , where  $\mathbb{B}_{W/2}(p)$  is the disk of radius  $W/2$  centered at  $p$ . The  $W$ -clearance condition is then given by:

$$\exists \mathcal{P}_V^W \subset \mathcal{F}_W(t) : \mathbf{x}_V^S \rightsquigarrow \mathbf{x}_V^G, \quad (2)$$

which certifies a collision-free centerline path for any vehicle footprint contained in the enclosing disk.

### C. Collaborative Pushing Modes

To enlarge blocked gaps, the robots may actively reconfigure  $\Omega$  by pushing movable obstacles. For obstacle  $\Omega_m \in \Omega$ , a pushing mode is denoted by  $\xi_m \triangleq (\mathbf{c}_m, \mathbf{u}_m)$ , where  $\mathbf{c}_m = (c_1, \dots, c_N)$  is an ordered tuple of contact points on  $\partial\Omega_m$ , and the ordering specifies the robot-to-contact assignment. The vector  $\mathbf{u}_m \in \mathbb{R}^{2N}$  stacks the corresponding planar contact forces, expressed in the body frame of  $\Omega_m$ . For a contact tuple  $\mathbf{c}_m$ , the set of admissible forces  $U_m(\mathbf{c}_m) \subset \mathbb{R}^{2N}$  collects contact forces satisfying the robot pushing limits, robot-object friction, and unilateral contact constraints. The body wrench induced by a mode is defined as:

$$\mathbf{q}_m(\xi_m) \triangleq G_m(\mathbf{c}_m)\mathbf{u}_m, \quad (3)$$

where  $G_m(\mathbf{c}_m) \in \mathbb{R}^{3 \times 2N}$  is the contact-to-wrench mapping; and  $\mathbf{q}_m(\xi_m) \in \mathbb{R}^3$ . The admissible mode set is state dependent as denoted by  $\Xi_m(\mathbf{s})$ . It contains modes whose contacts are reachable from the current robot configurations, pre-contact configurations are feasible in the current clutter, and forces satisfy  $\mathbf{u}_m \in U_m(\mathbf{c}_m)$ . Hence, applying a feasible mode  $\xi_m \in \Xi_m(\mathbf{s}(t))$  yields the physical transition:

$$\mathbf{s}(t^+) \triangleq \Phi(\mathbf{s}(t), m, \xi_m), \quad (4)$$

where  $\mathbf{s}(t)$  stacks all robot and obstacle states. *The force vector  $\mathbf{u}_m$  serves as a feasibility variable to certify wrench generation under the modeled contact and actuation constraints rather than a directly commanded control input.*

### D. Problem Statement

The goal is to compute a *hybrid schedule* that reconfigures movable obstacles to admit a  $W$ -feasible path for the external vehicle. The schedule is defined as:

$$\pi \triangleq \{(\Omega_{m_k}, \xi_k, \Delta t_k)\}_{k=1}^K, \quad (5)$$

where  $\Omega_{m_k} \in \Omega$  is the manipulated obstacle,  $\xi_k \in \Xi_{m_k}$  is its pushing mode, and  $\Delta t_k > 0$  is the execution duration. The search is guided by

$$\mathcal{J}_{\text{clr}}(\pi) \triangleq T + \alpha \sum_{k=1}^K J(m_k, \xi_k, s(t_k)), \quad (6)$$

where  $T \triangleq \sum_{k=1}^K \Delta t_k$  denotes the robotic clearance duration excluding subsequent vehicle traversal, and  $J(\cdot)$  is the simulation-estimated manipulation effort. Vehicle traversal is considered only through the terminal  $W$ -clearance condition (2). The constraints (1)–(4) enforce valid pushing transitions. Equation (6) ranks feasible schedules; PIHS uses finite sampling and does not guarantee global optimality of the continuous hybrid objective.

**Remark 1.** The objective and constraints above couple obstacle selection, pushing-mode generation, and switching in one hybrid planning problem, yielding a combinatorial-continuous search space far richer than classical NAMO formulations with simple objects or hand-crafted contact modes [8], [10], [11]. ■

### III. PROPOSED SOLUTION

As shown in Fig. 2, PushAround first uses the  $W$ -Clearance Connectivity Graph in Sec. III-A and gap ranking in Sec. III-B to identify and prioritize blocking gaps. It then generates and evaluates physically feasible collaborative pushing modes in Sec. III-C. These components are integrated into the simulation-in-the-loop hybrid search in Sec. III-D, followed by the online execution procedure summarized in Sec. III-E.

#### A. $W$ -Clearance Connectivity Graph

The feasibility of routing the vehicle reduces to whether a  $W$ -width corridor connects  $\mathbf{x}_v^S$  and  $\mathbf{x}_v^G$ . A  $W$ -clearance connectivity graph (WCCG) is introduced here to encode obstacle adjacencies under clearance  $W$  and enable fast connectivity queries. **Constructed directly from obstacle geometry rather than a workspace grid, the WCCG does not introduce a grid-resolution parameter in the repeated connectivity queries used during planning.**

1) *Graph Construction:* The WCCG is built by decomposing all clearance-relevant obstacle and workspace boundaries into convex components or segments. From each component  $C$ , a centroid node  $v_c$  is created. When two components  $C_u$  and  $C_v$  have closest points  $p_u \in C_u$  and  $p_v \in C_v$  with distance less than  $W$ , bridge nodes are added at  $p_u$  and  $p_v$ . These bridge nodes are connected by a bridge-bridge edge, annotated with the gap width  $w_{uv} \triangleq \|p_u - p_v\|$ , and further linked back to their centroids with centroid-bridge edges. Narrow passages with  $w_{uv} < W$  are marked as potential bottlenecks. The resulting graph is given by:

$$\mathcal{G}_W \triangleq (\mathcal{V}, \mathcal{E}_c \cup \mathcal{E}_b), \quad (7)$$

where  $\mathcal{V}$  are centroid and bridge nodes;  $\mathcal{E}_c$  contains centroid-bridge edges; and  $\mathcal{E}_b$  for bridge-bridge edges with widths  $w_{uv}$ .

2) *Connectivity Criterion:* Once  $\mathcal{G}_W$  has been constructed, connectivity queries can be performed without explicitly computing a geometric path. Let  $\text{Face}(\mathcal{G}_W)$  denote the set of connected planar faces induced by the embedded WCCG, i.e., the maximal regions separated by obstacle components and WCCG bottleneck edges. The criterion for (2) is given by:

$$(\mathbf{x}_v^S, \mathbf{x}_v^G \in \mathcal{F}_W(t)) \text{ and } (\exists F \in \text{Face}(\mathcal{G}_W) : \mathbf{x}_v^S, \mathbf{x}_v^G \in F), \quad (8)$$

The criterion follows the standard configuration-space inflation interpretation: expanding obstacles by a disk of radius  $W/2$  turns gaps narrower than  $W$  into overlapping barriers. Non-convex obstacles are handled through clearance-relevant decomposition, after which WCCG records the separating bottlenecks. The consistency between WCCG connectivity and continuous and rasterized configuration-space references is further evaluated in Sec. IV-A3. A frontier-tracing procedure, similar to the BugPlanner [20], is then used to extract a skeleton  $\Sigma$  as an ordered sequence of centroid and bridge nodes. Starting from  $\mathbf{x}_v^S$ , it casts a ray toward  $\mathbf{x}_v^G$  and follows the encountered frontier loop until either a valid exit is found or a blocking cycle is returned.

#### B. Ranking of Frontier Gaps

When the condition in (8) fails, the vehicle cannot reach  $\mathbf{x}_v^G$  from  $\mathbf{x}_v^S$  through a  $W$ -clear path  $\mathcal{P}_v^W$ . In this case, the planner must prioritize *blocking gaps* on the reachable frontier of  $\mathcal{G}_W$  according to their predicted downstream cost, and feed the ranked list to the hybrid search as local clearance targets.

1) *Frontier Extraction and Candidate Gaps:* A BugPlanner query on  $\mathcal{G}_W$  above either confirms connectivity or returns a counter-clockwise frontier loop  $\mathcal{L}$  that separates the two endpoints. The bridge-bridge edges on  $\mathcal{L}$  define the first-hop candidate set  $\Gamma_{\mathcal{L}} \triangleq \{g_1, \dots, g_K\}$ , where each gap  $g_k$  is represented by its two closest boundary points, its width  $w_{g_k} < W$ , and the adjacent obstacles that can potentially be moved. These candidates are then ranked by predicted downstream cost and passed to the hybrid search as local clearance targets.

2) *Evaluation for Immediate Cost:* Each candidate  $g \in \Gamma_{\mathcal{L}}$  is evaluated by combining the cost for the robots to reach the gap and the effort required to widen it. Let  $\mathbf{s}_{\mathcal{R}}$  be the current robot positions, and  $\mathbf{o}_g$  as the outside insertion point of gap  $g$ . The resulting one-hop cost is given by:

$$\mathcal{C}(g \mid \mathcal{L}, \mathbf{s}_{\mathcal{R}}) = \lambda_t \mathcal{C}_t(\mathbf{s}_{\mathcal{R}}, \mathbf{o}_g) + \lambda_p (\mathcal{C}_{\text{dp}}(g) + \mathcal{C}_{\text{rp}}(g)) \quad (9)$$

where  $\lambda_t, \lambda_p > 0$  weight transition and pushing costs, respectively;  $\mathcal{C}_t$ ,  $\mathcal{C}_{\text{dp}}$ , and  $\mathcal{C}_{\text{rp}}$  denote transition, direct-pushing, and recursive-pushing costs.

3) *Long-term Cost w.r.t. Goal:* Furthermore, to prioritize gaps closer to the goal, a heuristic is added to the evaluation, i.e.,  $h(g) = \eta \|\mathbf{o}(g) - \mathbf{x}_v^G\|$ , with the scaling constant  $\eta > 0$ . Lastly, a short A\*-style presearch is adopted to cross each candidate gap, recompute the next frontier, and continue for a limited width and depth. The *cost-to-connect* is predicted by:

$$\widehat{\text{Cost}}(g) \triangleq \mathcal{C}(g \mid \mathcal{L}, \mathbf{s}_{\mathcal{R}}) + \sum_{g' \in \Pi^*(g)} \{\mathcal{C}(g' \mid \cdot) + h(g')\}, \quad (10)$$

where  $\Pi^*(g)$  is the sequence of subsequent gaps discovered after virtually crossing  $g$ . Consequently, the final output of this

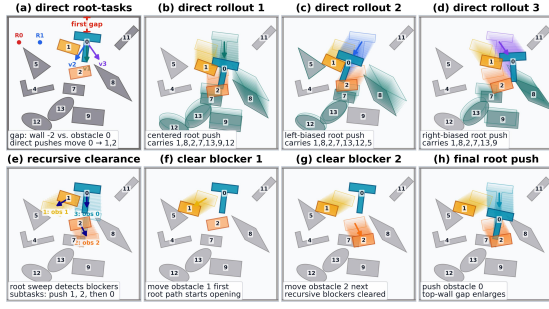


Fig. 3. **Direct** pushing moves a gap-side blocker along a gap-opening direction, whereas **recursive** pushing first clears movable blockers in the swept region and then returns to the root gap-opening task.

module is the ranked list of candidate gaps, i.e.,  $\text{Rank}(\Gamma_{\mathcal{L}}) \triangleq \text{argsort}_{g \in \Gamma_{\mathcal{L}}} \widehat{\text{Cost}}(g)$ , in ascending predicted cost. This ordered set is passed to the hybrid search module, such that only the promising gaps are expanded first.

### C. Pushing Mode Generation and Evaluation

For a selected frontier gap  $g$ , this subsection constructs a set of candidate pushing strategies  $\Xi_g \triangleq \{(\Omega_m, \mathbf{v}, \xi)\}$ , where  $\mathbf{v} \triangleq (v_x, v_y, \omega) \in \mathbb{R}^3$  is a short-horizon pushing direction for the movable obstacle  $\Omega_m$  induced by  $g$ , and  $\xi \triangleq (\mathbf{c}_m, \mathbf{u})$  is the associated pushing mode in (4).

1) *Direct and Recursive Pushing Tasks*: For *direct push*, let  $\Omega_m$  denote one of the two movable obstacles adjacent to  $g$ . For each such obstacle, a small set of candidate body velocities  $\{\mathbf{v}_m^k\}_{k=1}^{K_m}$  is sampled with a bias toward enlarging the gap  $g$ , while a few angular components are included to account for local geometry. However, if the swept region of a direct push intersects movable blockers, *recursive push* tasks are further required. Namely, for each out-layer blocker  $\Omega_m$ , the nominal direction moves it away from both the swept region and the reference gap sequence  $\Pi^*(g)$ , with the candidate velocities  $\{\mathbf{v}_m^k\}_{k=1}^{K_m}$ . The recursive pushes aim to move the blocker out of the swept region rather than directly open the gap. Recursion therefore creates a bounded dependency chain which prioritizes the final blocker  $\Omega_m$ , as shown in Fig. 3.

2) *Pushing Mode Generation*: At a search state  $s$ , a pushing mode is generated for each candidate direction  $\mathbf{v}_m^k$  and obstacle  $\Omega_m$ . In this subsubsection, contact tuples, object velocities, forces, and wrenches are expressed in the body frame of  $\Omega_m$ . For a contact tuple  $\mathbf{c}$ , its multi-directional force feasibility is:

$$J_{\text{MF}}^m(\mathbf{c}, \mathbf{v}_m^k) \triangleq \sum_{\mathbf{v}_d \in \mathcal{D}_m^k} w_d \min_{\mathbf{u} \in U_m(\mathbf{c})} \|\mathbf{q}_m(\mathbf{c}, \mathbf{u}) + \mathbf{q}_m^\mu(\mathbf{v}_d)\|, \quad (11)$$

where  $\|\cdot\|$  is a weighted  $\ell_1$  wrench norm, the finite set  $\mathcal{D}_m^k$  perturbs  $\mathbf{v}_m^k$ , and  $\mathbf{q}_m^\mu(\mathbf{v}_d)$  is the quasi-static friction wrench; the inner minimization is a small LP over the admissible forces  $U_m(\mathbf{c})$ . A state- and direction-dependent contact set  $\mathcal{A}_m^k(s) \subset \partial\Omega_m$  is drawn from the robot-reachable boundary of  $\Omega_m$  by velocity-force compatibility filtering (Fig. 4), keeping a contact only if its robots reach collision-free pre-contact poses. Starting from cached layouts or random initializations, the tuple is refined by sequential greedy replacement: at the

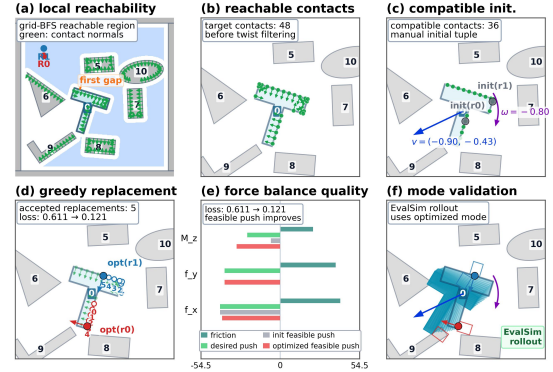


Fig. 4. Reachable contacts are filtered by the desired object twist, assigned to robots, refined by local replacement, and finally validated by simulation.

$\ell$ -th step,  $\mathbf{c}_{r,c}^\ell$  replaces the  $r$ -th contact with  $c$ , and the best replacement is:

$$c^* = \underset{c \in \mathcal{A}_m^k(s)}{\text{argmin}} J_{\text{MF}}^m(\mathbf{c}_{r,c}^\ell, \mathbf{v}_m^k), \quad (12)$$

swept over contact slots until  $J_{\text{MF}}^m$  no longer decreases; a bounded backtracking variant removes the robot-order sensitivity of this replacement. If no full-team assignment exists for a task, e.g., on small faces or under the pairwise pusher-clearance constraint, one robot is released for that single push (charged a penalty when ranking modes, so full-team modes are preferred) and the full team is available again afterwards. The best tuple over initializations gives  $\mathbf{c}_m^{k,*}$ , from which  $\mathbf{u}_m^k$  follows from the inner minimization in (11), yielding  $\xi_m^k = (\mathbf{c}_m^{k,*}, \mathbf{u}_m^k)$ ; each mode is cached as a warm start. The best few admissible pairs are kept for gap  $g$  and passed to simulation;  $\mathbf{u}_m^k$  certifies wrench feasibility while hardware tracks the contact geometry from pose/velocity commands.

3) *Pushing Strategy Evaluation*: For a current state  $s$  and a candidate strategy  $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$ , a short-horizon rollout evaluates the transition in (4) under the selected mode. As shown in Fig. 4, a strategy is accepted only if the robots reach the pre-contact poses, maintain stable contact, and induce obstacle motion above threshold. The simulation returns:

$$(s', \delta_T, \delta_J) \triangleq \text{EvalSim}(s, (\Omega_m, \mathbf{v}, \xi)), \quad (13)$$

where  $s'$  is the successor state capturing the resulting robot and object configurations;  $\delta_T > 0$  is the time cost; and  $\delta_J$  is the control effort. The rollout jointly simulates all robots and movable bodies, thereby capturing contact-induced object interactions. For efficiency, the pushing rollout may be bypassed when the mode has been validated before and the swept region of  $\Omega_m$  is collision-free.

### D. Physics-Informed Hybrid Search

The proposed physics-informed hybrid search (PIHS) selects blocking gaps and validates corresponding multi-robot pushing strategies through short-horizon simulation, as shown in Fig. 5 and summarized in Alg. 1. Unlike purely geometric NAMO planners, PIHS expands only physics-validated strategies through parallel candidate evaluation. It is a feasibility-oriented heuristic search, where Eq. (6) defines the ranking objective for candidate schedules. The search operates on a finite sampled candidate set and therefore does not guarantee

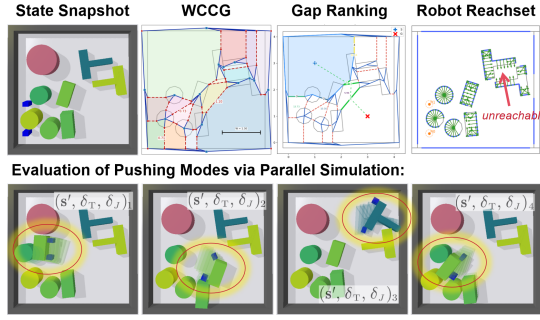


Fig. 5. **Top:** Core components of the proposed physics-informed hybrid search; **Bottom:** Sampled push tasks are evaluated via parallel simulation, producing updated states for search expansion.

global optimality or completeness of the underlying continuous hybrid planning problem.

1) *Tree and Initialization:* The search tree  $\mathcal{T}$  is composed of nodes  $\nu \triangleq (s, \pi)$ , where  $s$  is the current system state of all robots and movables in (4), and  $\pi$  the partial pushing strategy in (6). Two global functions are maintained:  $\text{Rank}(\nu)$  stores ranked candidate gaps  $\Gamma_{\mathcal{L}_\nu}$  given the frontier loop  $\mathcal{L}_\nu$  associated with node  $\nu$ , together with their exploration costs by (10); and  $\chi(\nu)$  returns the cumulative execution cost from the root to  $\nu$ . The root is initialized as  $\nu_0 \triangleq (s_0, \emptyset)$ , where  $s_0$  is the initial system state.

2) *Node Selection:* At each iteration, the node with minimum best-first priority is selected from the queue. Priority balances realized cost and estimated remaining effort:

$$f(\nu) \triangleq \chi(\nu) + \min_{g \in \text{Rank}(\nu)} \widehat{\text{Cost}}(g), \quad (14)$$

where  $\chi(\nu)$  is the cumulative execution cost so far, and the second term is the minimum predicted remaining cost among the unexplored gaps of  $\nu$ . The remaining-cost term is a heuristic and is not required to be admissible. The score prioritizes nodes by realized cost and promising future actions, but it does not provide an A\* optimality guarantee.

3) *Node Expansion with Parallel Simulation:* Let the selected node be  $\nu \triangleq (s, \pi)$ . For each ranked gap  $g \in \text{Rank}(\nu)$ , its predicted gap sequence  $\Pi^*(g)$  in (10) is first derived. Then, the mode optimization by (11) provides a candidate strategy set  $\Xi_g \triangleq \{(\Omega_m, \mathbf{v}, \xi)\}$ , including both the direct push of the adjacent movable obstacles, and recursive push of out-layer obstacles. Thus, expansion is prioritized first by the gap ordering in  $\text{Rank}(\nu)$ , and then by the candidate ordering within  $\Xi_g$ . Each candidate strategy  $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$  is first checked by a geometric quick-pass and, if it passes, evaluated by the short-horizon rollout in (13). A successful evaluation yields  $(s', \delta_T, \delta_J)$ , from which a child node  $\nu' \triangleq (s', \pi')$  is created by appending  $\tau \triangleq (\Omega_m, \mathbf{v}, \xi)$  to the current partial schedule  $\pi$ . The cumulative cost is updated by:  $\chi(\nu') \triangleq \chi(\nu) + \widehat{C}(\nu, \tau)$ , where  $\widehat{C}(\nu, \tau) \triangleq \delta_T + \alpha \delta_J$  is the realized incremental cost. Note that all candidate strategies in the current batch are evaluated *concurrently in parallel*, and a single expansion can produce multiple validated successors.

4) *Termination:* Termination occurs when a node  $\nu$  reaches a state  $s$  for which a  $W$ -clear path  $\mathcal{P}_V^W$  exists from  $\mathbf{x}_V^S$  to  $\mathbf{x}_V^G$  by (8). Thus, the schedule  $\pi$  stored in  $\nu$  is a feasible solution satisfying the terminal clearance condition and the validated

### Algorithm 1: Physics-Informed Hybrid Search

---

**Input:**  $s_0, \alpha, \text{EvalSim}(\cdot), W$ -criterion by (8)  
**Output:**  $\pi^*$

```

1  $\nu_0 \leftarrow (s_0, \emptyset), \chi(\nu_0) = 0;$ 
2 Init  $\mathcal{Q}$  by (14);
3 while not terminated do
4    $\nu \leftarrow \mathcal{Q}.\text{pop\_min}()$  by (14);
5   foreach  $g \in \text{Rank}(\nu)$  do
6     Compute  $\Pi^*(g)$ , generate  $\Xi_g$ ;
7     foreach  $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$  do
8        $\tau = (\Omega_m, \mathbf{v}, \xi)$  // Direct or recur.
9        $(s', \delta_T, \delta_J) \leftarrow \text{EvalSim}(\nu, \tau)$  by (13);
10      if success then
11         $\nu' \leftarrow (s', \pi \cup \{\tau\});$ 
12         $\chi(\nu') \leftarrow \chi(\nu) + \delta_T + \alpha \delta_J;$ 
13        Add  $\nu'$  to  $\mathcal{Q}$ ;
14      if  $s'$  admits  $\mathcal{P}_V^W$  by (8) then
15         $\pi^* \leftarrow \pi', \text{break};$ 
16 return  $\pi^*;$ 

```

---

pushing transitions. It encodes both the sequence of gaps to clear and the pushing strategies that realize them, without implying that  $\mathcal{J}_{\text{clr}}(\pi)$  is globally minimal.

**Remark 2.** Efficiency comes from three lightweight heuristics: gap-lookahead reduces unproductive branches, MODE-CACHE warm-starts mode optimization, and the quick-pass rollout bypass skips simple motions whose mode was previously validated, avoiding redundant simulations. ■

### E. Overall Analyses

1) *Execution and Online Adaptation:* The hybrid search returns a pushing schedule  $\pi = \{\tau_1, \dots, \tau_K\}$ , with  $\tau_k \triangleq (\Omega_k, \mathbf{v}_k, \xi_k)$ . The schedule is executed in a receding-horizon manner: for each  $\tau_k$  the robots reach the mode  $\xi_k$  and follow the pushing direction  $\mathbf{v}_k$  from the current state, as in (13). On the hardware the optimized contact forces certify feasibility rather than being commanded; the controller converts the planned contact geometry and object motion into velocity commands with feedback tracking. During transitions, robots are reassigned by preserving the circular order of contact points along the obstacle boundary [7], reducing path crossings under sufficient clearance. After each push the state is updated from observation, the WCCG is rebuilt from the current configuration, and execution continues until a  $W$ -clear path  $\mathcal{P}_V^W$  connects  $\mathbf{x}_V^S$  and  $\mathbf{x}_V^G$ . Repanning is triggered by observed execution discrepancies – failed contact establishment, stalled pushing, or invalidated WCCG connectivity – in which case Alg. 1 is called from the current state to derive the updated schedule.

2) *Computational Complexity:* Given  $M$  movable obstacles, the computational cost of one node expansion in Alg. 1 is dominated by candidate generation and physics validation. The WCCG update and gap ranking incur  $\mathcal{O}(M \log M + |\mathcal{V}| + |\mathcal{E}_c| + |\mathcal{E}_b|)$  and  $\mathcal{O}(|\Gamma_{\mathcal{L}_\nu}| \log |\Gamma_{\mathcal{L}_\nu}|)$  costs, respectively, while the dominant simulation cost is

$$\mathcal{O} \left( \sum_{g \in \text{Rank}(\nu)} |\Xi_g| \left( C_{\text{LP}} + \frac{\rho_\nu T_{\text{sim}}}{n_w} \right) \right),$$

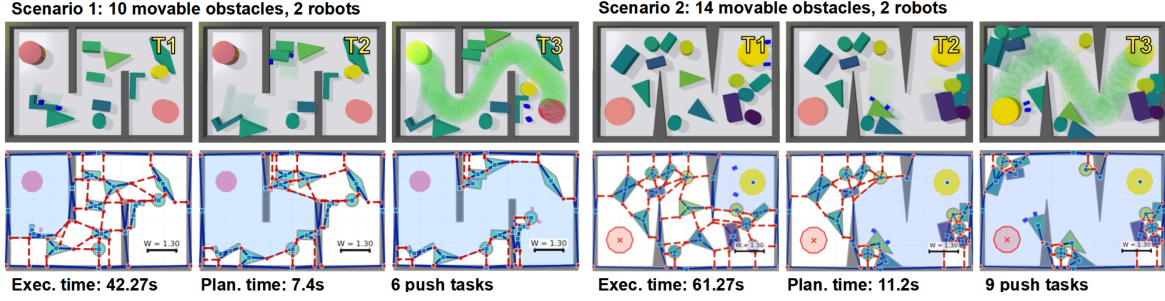


Fig. 6. Execution and planning alignment in Scenario 1 (Left) and Scenario 2 (Right). **Top:** simulated snapshots during path clearing. **Bottom:** corresponding WCCG overlays with the start face in blue. The executed pushes progressively expand the reachable  $W$ -clearance region until the goal is connected.

TABLE I  
WCCG CONNECTIVITY VALIDATION AND MEAN RUNTIME.

| Type       | CC=FR   | FP/FN <sup>†</sup> | Time (ms) |       |             |
|------------|---------|--------------------|-----------|-------|-------------|
|            |         |                    | CC        | FR    | WCCG        |
| Convex     | 100/100 | 0/0                | 1.23      | 282.9 | <b>0.86</b> |
| Non-convex | 98/100  | 0/0                | 1.51      | 204.1 | <b>1.30</b> |
| Mixed      | 98/100  | 0/0                | 1.33      | 242.4 | <b>1.05</b> |
| Total      | 296/300 | <b>0/0</b>         | 1.36      | 243.1 | <b>1.07</b> |

<sup>†</sup> WCCG errors on CC-FR-consistent cases. All runtimes are measured in one harness over the same 300 scenes, with WCCG served by its compiled backend;

Therefore, scalability is mainly determined by validated pushing candidates, while quick-pass filtering and parallel simulation reduce the practical expansion cost. Robot-team size mainly affects mode generation, where greedy contact refinement avoids exhaustive  $|A|^N$  contact assignment and evaluates bounded single-contact replacements.

#### IV. EXPERIMENTS

Numerical experiments are conducted to validate the proposed PIHS algorithm. The algorithms are implemented in Python3 and PyBullet [21] on a computer with an Intel Core i9-14900KF and an NVIDIA RTX 4080. Videos are provided as multimedia attachments.

##### A. Numerical Simulations

1) *Setup and Configuration:* The simulation uses a step size of  $\Delta t = 1/240$  s and a control period of  $1/40$  s. Robots are modeled as disk or box pushers consistent with the  $W$ -clearance definition, while movable obstacles have masses uniformly sampled from  $[5, 15]$  kg. A trial succeeds when the external vehicle reaches its goal through a  $W$ -clear path. The nominal scenario uses an  $8 \times 5$  m workspace with bottleneck-forming bar obstacles and randomly distributed movable objects. Two robots clear 10–15 movable obstacles under minimum-separation constraints. Unless otherwise stated, comparisons and ablations are conducted on Scenario 1 with 10 movable obstacles over 40 trials, while Scenario 2 with 14 obstacles provides an additional cluttered case. All methods use the same robot footprints, obstacle configurations, collision/reachability checks, and execution simulator, and are given the same per-trial timeout of 800 s and planning budget of 20,000 control steps; unsuccessful trials are assigned the 800 s cap in the planning-time statistics.

2) *Results:* Two representative simulation runs are shown in Fig. 6. In Scenario 1 (10 movable obstacles, two robots) the start and goal belong to different  $W$ -clearance faces, so the planner clears the blocked passage with 6 pushes in 7.4 s and

TABLE II  
PERFORMANCE COMPARISON AND ABLATIONS ON SCENARIO 1 OVER 40 RANDOMIZED TRIALS.

| Method / Variant         | Succ.(%)    | PT (s)      | ET (s)      | #Sims        | #Pushes    |
|--------------------------|-------------|-------------|-------------|--------------|------------|
| DFS-WCCG                 | 25.0        | >100.0      | 51.2        | >500         | 8.0        |
| SL-Push (offline)        | 62.5        | 0.06        | 44.1        | 0.0          | 7.3        |
| SL-Push (sim)            | 75.0        | 18.2        | 64.6        | 20.0         | 10.2       |
| Rec-NAMO                 | 37.5        | 13.3        | 42.4        | 0.0          | 7.8        |
| <b>PushAround (ours)</b> | <b>97.5</b> | <b>10.3</b> | <b>28.6</b> | <b>121.5</b> | <b>6.0</b> |
| w/o gap-lookahead        | 82.5        | 16.9        | 35.2        | 178.3        | 7.2        |
| w/o mode-cache           | 92.5        | 17.1        | 33.4        | 164.6        | 6.5        |
| w/o quick-pass           | 90.0        | 23.6        | 41.9        | 208.2        | 6.1        |
| w/o reinsertion          | 85.0        | 12.8        | 31.1        | 151.7        | 7.8        |



Fig. 7. Failure cases by baselines Rec-NAMO, SL-Push and DFS-WCCG.

the execution finishes in 42.27 s with the start face containing the goal. Scenario 2 (14 obstacles) requires a 9-push plan in 11.2 s and 61.27 s of execution. In both cases the executed pushes progressively enlarge the reachable  $W$ -clearance region until a complete  $W$ -clear path is found, as shown by the expanding blue start face in the WCCG overlays.

3) *WCCG Connectivity Validation:* WCCG is compared against continuous  $C$ -space (CC) and a rasterized reference at 0.02 m (FR) on 300 random scenarios (Table I): CC and FR agree in 296 cases, WCCG agrees with CC in all 300 with zero FP/FN, and a query costs 1.07 ms against 1.36 ms (CC) and 243 ms (FR), while additionally supplying the gap graph needed by PIHS.

4) *Comparisons:* Four baselines are evaluated on Scenario 1 over 40 randomized trials under the same  $W$ -clearance criterion, object distribution, and simulator: **DFS-WCCG** (WCCG-guided search with fixed pushes), **SL-Push** (offline and simulation-validated variants), and **Rec-NAMO** (recursive NAMO) [3], [4], [16], [5]. Table II shows that PushAround reaches the highest success rate (97.5%) and the lowest combined planning-and-execution time (38.9 s). DFS-WCCG needs more than 100 s of planning (exponential branching); SL-Push (offline) plans in 0.06 s but many displacements fail in execution (62.5%); SL-Push (sim) reaches 75.0% at 18.2 s

TABLE III  
SCALABILITY AND ROBUSTNESS UNDER OBSTACLE DENSITY,  
ROBOT-TEAM SIZE, AND EXECUTION-SIDE PHYSICS MISMATCH.

| Case                                                           | Trials | Succ.(%) | PT (s)            | ET (s)            |
|----------------------------------------------------------------|--------|----------|-------------------|-------------------|
| <i>Obstacle density</i> (PushAround, $N=2$ , nominal)          |        |          |                   |                   |
| $M = 15$                                                       | 10     | 100      | $2.5 \pm 0.9$     | $37.1 \pm 21.0$   |
| $M = 30$                                                       | 10     | 100      | $11.7 \pm 3.7$    | $100.9 \pm 34.7$  |
| $M = 45$                                                       | 10     | 100      | $53.4 \pm 24.5$   | $272.7 \pm 97.4$  |
| <i>Robot-team size</i> ( $M=30$ , nominal; 10 trials per cell) |        |          |                   |                   |
| PushAround, $N=3$                                              | 10     | 100      | $30.0 \pm 12.7$   | $97.3 \pm 42.1$   |
| PushAround, $N=4$                                              | 10     | 100      | $29.2 \pm 8.9$    | $110.1 \pm 33.1$  |
| SL-Push, $N=3$                                                 | 10     | 80       | $30.0 \pm 46.3$   | $193.3 \pm 111.3$ |
| SL-Push, $N=4$                                                 | 10     | 80       | $23.4 \pm 18.2$   | $171.7 \pm 61.2$  |
| DFS-WCCG, $N=3$                                                | 10     | 100      | $142.6 \pm 183.1$ | $178.1 \pm 72.1$  |
| DFS-WCCG, $N=4$                                                | 10     | 90       | $101.9 \pm 103.9$ | $175.9 \pm 113.6$ |
| <i>Physics mismatch</i> ( $M=30$ , $N=2$ )                     |        |          |                   |                   |
| Nominal                                                        | 10     | 100      | $16.8 \pm 3.4$    | $104.7 \pm 37.4$  |
| Mass $\times 0.6-1.4$                                          | 40     | 100      | $16.8 \pm 5.4$    | $105.9 \pm 44.6$  |
| Lateral fric. $\times 0.6-1.4$                                 | 40     | 97.5     | $16.5 \pm 5.7$    | $114.0 \pm 48.5$  |
| Spin fric. $\times 0.6-1.4$                                    | 40     | 97.5     | $15.2 \pm 3.6$    | $100.1 \pm 35.7$  |
| Force limit $\times 0.6-1.4$                                   | 40     | 95.0     | $18.9 \pm 10.9$   | $115.9 \pm 58.5$  |
| All perturbed                                                  | 160    | 97.5     | $16.9 \pm 7.0$    | $108.9 \pm 47.4$  |

Each perturbed physics row scales one execution-side parameter over the factor range shown while the others remain nominal.

planning and 10.2 pushes; Rec-NAMO often fails to build a connected  $W$ -clear corridor in dense clutter. Figure 7 shows representative baseline failures.

5) *Ablation Studies*: Four internal ablations are conducted: **w/o gap-lookahead** (frontier gaps ranked without downstream costs), **w/o mode-cache** (no warm start from previously validated modes), **w/o quick-pass** (all candidates evaluated by rollout), and **w/o reinsertion** (temporarily failed nodes not reconsidered). Table II shows that gap-lookahead and reinsertion mainly improve robustness (success drops by more than 10% when either is removed), whereas mode-cache is a warm start and only lowers success to 92.5%. Quick-pass affects efficiency: without it simulations rise from 121.5 to 208.2 and planning from 10.3 to 23.6 s.

6) *Scalability and Robustness Analysis*: Generalization is further evaluated in the same  $8 \times 8$  m workspace by scaling the obstacle density  $M$ , the robot-team size  $N$ , and the execution-side physics parameters; Table III summarizes representative settings and Fig. 9 plots the complete density sweep. (1) **Obstacle density**: with two robots and 10 trials per density, PushAround keeps a 100% success rate for all  $M \in [15, 45]$  while its mean planning time grows from 2.5 to 53.4 s, whereas all baselines solve none of the  $M=45$  trials and removing recursive pushes lowers success to 70% (Fig. 8 shows representative runs). (2) **Robot-team size**: at  $M=30$ , PushAround solves all ten trials at  $N=2, 3$ , and 4, after two mode-generation refinements: bounded backtracking contact assignment removes the robot-order sensitivity of greedy replacement, and when no full-team assignment exists for a task, one robot is released for that single push and re-activated afterwards, with a penalty favoring full-team modes. Since the fallback belongs to the shared mode generator, the baselines benefit as well (Table III); PushAround is nevertheless the only method that solves all ten trials at every team size and the most economical, e.g., at  $N=4$  with 29.2 s of planning, 110.1 s of execution and 10 pushes versus 101.9 s, 175.9 s and 14 for DFS-WCCG, and 171.7 s of execution and 20 pushes

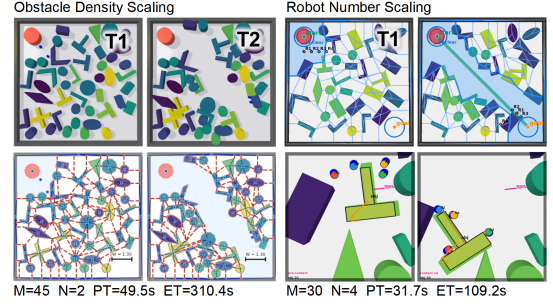


Fig. 8. Scalability of the clearing process. **Left**: obstacle-density scaling ( $M=45$  objects, two robots): snapshots during the clearing (**top**) and WCCG overlays with the start face in blue (**bottom**). **Right**: robot-team scaling ( $M=30$  objects, four robots): initial and final states (**top**) and zoomed views of one push before contact and during execution (**bottom**).

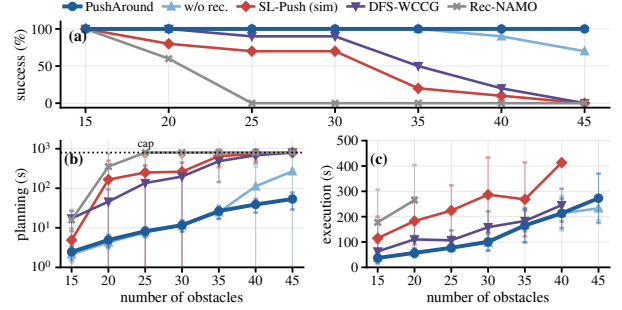


Fig. 9. Obstacle-density scaling ( $8 \times 8$  m, 10 independent seeds per density). (a) Success rate; (b) mean planning time, with unsuccessful trials capped at the 800 s budget (dotted line); (c) mean execution time of the successful trials (Rec-NAMO has none beyond  $M=20$ ). Error bars denote one standard deviation.

for SL-Push (sim). (3) **Physics mismatch**: with planning on the nominal model and only the execution simulator perturbed (one parameter at a time), PushAround succeeds in 156 of 160 perturbed trials (97.5%; nominal 10/10), the four failures being one low-friction planning failure and three budget time-outs. Under the same  $\pm 40\%$  perturbations the baselines are far less robust: SL-Push (sim) averages 32.5% success (0% under the strongest friction and force reductions) and DFS-WCCG 57.5% at an order of magnitude higher planning time.

## B. Hardware Experiments

1) *System Description*: Experiments use a  $5 \times 6$  m workspace of interlocking  $0.6 \times 0.6$  m foam mats (Fig. 10), 2 UGVs and 6 cardboard-box movable obstacles. Robots and movables each carry three to four motion-capture markers whose global poses are streamed to PyBullet in real time. The two Mecanum UGVs are controlled through planar velocity commands, while the optimized contact forces are used only for feasibility evaluation rather than direct motor control. Motion capture provides the current robot and object poses, which are compared with the successor state predicted by EvalSim: a push is deemed to deviate and triggers replanning only when the measured object position error exceeds 0.3 m or its orientation error exceeds  $\pi/4$ . The threshold is deliberately loose, tolerating a slight over-push or off-axis push. Movables are randomly placed for each trial.

2) *Results*: Two representative real-world runs are shown in Fig. 10. In Scenario 1, PIHS breaks five candidate gaps within 6.5 s using 12 node expansions, 49 visited snapshots

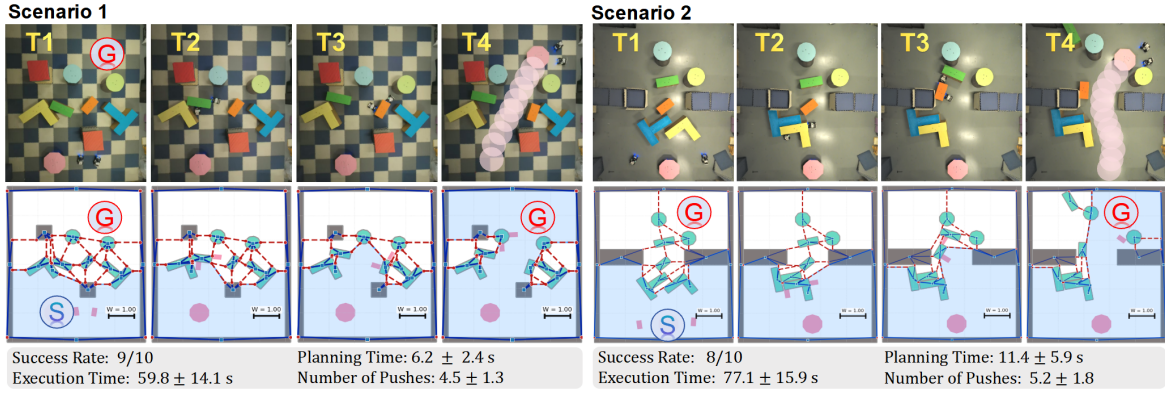


Fig. 10. Real-world pushing experiments with the execution–planning alignment (over 20 runs and 2 scenarios). **Top:** snapshots of two robots pushing obstacles in a  $5 \times 6$  m workspace with movable objects and fixed boundaries. **Bottom:** WCCG overlays with the start face in blue.

and search depth 6; the four-task schedule first pushes and rotates the yellow L-shaped object to open a gap near an immovable obstacle, then pushes the green and orange objects apart until the green cylinder is pushed to the right and a  $W$ -clear path is established at  $t = 52$  s. In Scenario 2, PushAround succeeds in 8/10 trials with an average of 5.2 pushes and 11.4 s of planning; it first selects a direct push on the blue L-shaped object, which also moves the adjacent T-shaped block and opens the first gap into the narrow passage. **This neighboring-object motion is not a separately encoded action but an emergent outcome of the coupled rigid-body rollout in EvalSim, whose snapshot is used to update the WCCG and continue replanning.** The complete execution takes 77.1 s on average with 15.9 s deviations, mainly from direction-dependent force limits, drifting, and Mecanum-wheel slippage near  $45^\circ$  motion directions, which are compensated by online replanning.

## V. CONCLUSION

This work presents a multi-robot framework PushAround for path clearing in unstructured environments using a hybrid search algorithm that jointly plans obstacles selection, contact points, and pushing forces efficiently. The method demonstrates hardware feasibility and uses state-feedback replanning to recover from observed execution deviations. Future directions include obstacle states estimation without external sensing, uncertain masses, and limited robot visibility for real-world applications.

## REFERENCES

- [1] L. Liu, X. Wang, X. Yang, H. Liu, J. Li, and P. Wang, "Path planning techniques for mobile robots: Review and prospect," *Expert Systems with Applications*, vol. 227, p. 120254, 2023.
- [2] M. Stilman and J. J. Kuffner, "Navigation among movable obstacles: Real-time reasoning in complex environments," *International Journal of Humanoid Robotics*, vol. 2, no. 04, pp. 479–503, 2005.
- [3] M. Stilman, J.-U. Schamburek, J. Kuffner, and T. Asfour, "Manipulation planning among movable obstacles," in *Proceedings 2007 IEEE international conference on robotics and automation*, 2007, pp. 3327–3332.
- [4] L. Yao, V. Modugno, A. M. Delfaki, Y. Liu, D. Stoyanov, and D. Kanoulas, "Local path planning among pushable objects based on reinforcement learning," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 3062–3068.
- [5] Z. Ren, B. Suvonov, G. Chen, B. He, Y. Liao, C. Fermuller, and J. Zhang, "Search-based path planning in interactive environments among movable obstacles," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 533–539.
- [6] J. J. Weeda, S. Bakker, G. Chen, and J. Alonso-Mora, "Pushing through clutter with movability awareness of blocking obstacles," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 512–518.
- [7] Z. Tang, Y. Zhang, and M. Guo, "Pushingbots: Collaborative pushing via neural accelerated combinatorial hybrid optimization," *IEEE Transactions on Robotics*, vol. 42, pp. 579–599, 2026.
- [8] S. Goyal, A. Ruina, and J. Papadopoulos, "Limit surface and moment function descriptions of planar sliding," in *IEEE International Conference on Robotics and Automation (ICRA)*, 1989, pp. 794–795.
- [9] K. M. Lynch, H. Maekawa, and K. Tanie, "Manipulation and active sensing by pushing using tactile feedback," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, vol. 1, 1992, pp. 416–421.
- [10] J. Chen, M. Gauci, W. Li, A. Kolling, and R. Groß, "Occlusion-based cooperative transport with a swarm of miniature mobile robots," *IEEE Transactions on Robotics*, vol. 31, no. 2, pp. 307–321, 2015.
- [11] Y. Wang and C. W. De Silva, "Multi-robot box-pushing: Single-agent q-learning vs. team q-learning," in *IEEE/RSJ international conference on intelligent robots and systems*, 2006, pp. 3694–3699.
- [12] Y. Feng, C. Hong, Y. Niu, S. Liu, Y. Yang, and D. Zhao, "Learning multi-agent loco-manipulation for long-horizon quadrupedal pushing," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 14 441–14 448.
- [13] Y.-C. Lin, B. Ponton, L. Righetti, and D. Berenson, "Efficient humanoid contact planning using learned centroidal dynamics prediction," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2019, pp. 5280–5286.
- [14] Q. Rouxel, S. Ivaldi, and J.-B. Mouret, "Multi-contact whole-body force control for position-controlled robots," *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 5639–5646, 2024.
- [15] T. Xue, H. Girgin, T. S. Lembono, and S. Calinon, "Demonstration-guided optimal control for long-term non-prehensile planar manipulation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 4999–5005.
- [16] D. M. Saxena and M. Likhachev, "Planning for complex non-prehensile manipulation among movable objects by interleaving multi-agent pathfinding and physics-based simulation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 8141–8147.
- [17] K. Ren, G. Wang, A. S. Morgan, L. E. Kavraki, and K. Hang, "Object-centric kinodynamic planning for nonprehensile robot rearrangement manipulation," *IEEE Transactions on Robotics*, vol. 41, pp. 5761–5780, 2025.
- [18] R. Ni and A. H. Qureshi, "Progressive learning for physics-informed neural motion planning," in *Robotics: Science and Systems*, 2023.
- [19] Y. Liu, R. Ni, and A. H. Qureshi, "Physics-informed neural mapping and motion planning in unknown environments," *IEEE Transactions on Robotics*, vol. 41, pp. 2200–2212, 2025.
- [20] K. N. McGuire, G. C. H. E. de Croon, and K. Tuyls, "A comparative study of bug algorithms for robot navigation," *Robotics and Autonomous Systems*, vol. 121, p. 103261, 2019.
- [21] E. Coumans and Y. Bai, "Pybullet, a python module for physics simulation for games, robotics and machine learning," <http://pybullet.org>, 2016–2019.